

WHICH ROTATIONS JOIN TWO MOSER LATTICES INTO A 5-CHROMATIC GRAPH?

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ABSTRACT. The smallest 5-chromatic unit-distance graphs known in the plane are of Parts’ *type M*: two subsets of the Moser lattice R glued at the origin by a single rotation. The record, 509 vertices since 2020, uses $\theta_4 = \arccos(7/8)$, and Parts writes that constructions of this type with other rotations “are not known yet”. Writing $\cos \theta_a = (2a - 1)/(2a)$, we prove that a joint lattice $R + \theta_a(R)$ of rank eight carries a homomorphism 4-colouring whenever $4 \nmid a$, and—once its shell is non-empty—only then, so a 5-chromatic $L \cup \theta_a(S)$ forces $4 \mid a$. The proof rests on one structural fact: $3R$ is the maximal order of $\mathbb{Q}(\sqrt{33}, i\sqrt{3})$. From it, the two integral geometric 4-colourings of R are the reductions modulo the two primes above 2, and shell occupancy becomes a norm condition in closed form; the integer candidates with an occupied shell have density zero. We exhibit constructions at six further rotations, three of them new— $\theta_{16}, \theta_{28}, \theta_{36}$ —while the three rational ones follow from Parts’ own monochromatic-pair gadgets, chaining of which gives the infinite family $a = 64n^2/9$ and with it $a = 64$, the one integer spindle our filters cannot decide. Five type-M rotations were in fact already known, only θ_4 among them a spindle. Finally, any 5-chromatic $L \cup \theta_4(S)$ needs at least five cross edges, and exactly 1920 five-edge sets defeat every homomorphism pairing; whether one is realised is open.

1. INTRODUCTION

The Hadwiger–Nelson problem asks for $\chi(\mathbb{R}^2)$, the least number of colours needed so that no two points at distance 1 share one. De Grey [1] proved $\chi(\mathbb{R}^2) \geq 5$ in 2018 with a 1581-vertex unit-distance graph; Polymath16, Heule [4] and Parts [6] drove the vertex count down to 509, where it has stood since 2020.

The record is built out of one lattice, and so is everything in this paper. Fix the *Moser lattice* R , the subgroup of $(\mathbb{C}, +)$ generated by the 30 unit vectors

$$\exp(i(k \cdot 60^\circ + j \theta_3/2)), \quad 0 \leq k < 6, \quad |j| \leq 2, \quad \theta_3 = \arccos \frac{5}{6},$$

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where $\theta_3 = \arccos \frac{5}{6}$ is the argument of $\omega_3 = \frac{5}{6} + \frac{\sqrt{11}}{6}i$ and 60° that of $\omega_1 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ (Figure 1). Those two do not by themselves give all thirty: ω_1 and ω_3 and their inverses generate only the eighteen with j even, and the twelve with j odd need the half-angle $\varphi = \frac{\sqrt{33}}{6} + \frac{\sqrt{3}}{6}i = e^{i\theta_3/2}$ as well—the same φ that generates the lattice in [9], met again in Section 6. The group is dense in the plane, and the reason is not that it has rank 4: it contains $\frac{1}{3}$ and $\frac{\sqrt{33}}{3}$, hence the subgroup $\frac{1}{3}\mathbb{Z} + \frac{\sqrt{33}}{3}\mathbb{Z}$, which is dense in the real axis because $\sqrt{33}$ is irrational; and it is carried to itself by the rotation through 60° , which supplies a second, independent line. A closed subgroup of $(\mathbb{C}, +)$ containing two independent dense lines is all of $\mathbb{C}$. Rank 4 alone would not give it: $(\mathbb{Z} + \sqrt{2}\mathbb{Z} + \sqrt{3}\mathbb{Z}) \times \mathbb{Z}$ has rank 4 and is not dense. Nor does $\varphi \in R$ make $\varphi R = R - R$ is no ring, and $\varphi^3 = (\sqrt{33} + 4i\sqrt{3})/9$ is already outside it.

A warning about the name, and then the fact that replaces it. R is a *group*, not a ring: it is not closed under multiplication— $\omega_3^2 = \frac{7}{18} + \frac{5\sqrt{11}}{18}i$ is not in it, and of the 900 products of pairs of unit vectors only 684 are. But $3R$ is a ring, and more than that: it is the maximal order of the field V that R spans, so

$$R = \frac{1}{3} \mathcal{O}_V, \quad V = \mathbb{Q}(\sqrt{33}, i\sqrt{3}),$$

which is Proposition 8 below and is what Sections 4 and 6 actually run on. The *Moser ring* that R generates is a strictly larger object, with denominators 3^k ; Gibbs' homomorphism and Hubai's 22 colourings [5] live there, while everything below is about R itself—as is Dúcz's uniqueness theorem [2] in spirit, though not literally: his lattice is the smaller one generated by $1, \omega_1, \omega_3, \omega_1\omega_3$, of index 9 in R , a difference Section 4 returns to. What R is closed under is multiplication by $\mathcal{O} = \mathbb{Z}[(1 + \sqrt{33})/2]$, and Section 4 runs on that alone.

Take $L, S \subseteq R$, rotate the second by a fixed θ , and glue them at the origin:

$$G = L \cup \theta(S).$$

Parts calls this *type M*, after Heule, and describes it as two base graphs rotated relative to each other so as to obtain many additional unit edges of two or more kinds. It is one of four types he classifies—A, M, J and T—and the one at issue here: type M, he writes, “is of the greatest interest, because so far it gives 5-chromatic graphs with the least number of vertices”. The only rotation he records for it is $\theta_4 = \arccos(7/8)$, which he writes as $\rho = \exp(i \arccos 7/8)$; writing in 2020 he could not have had the four that [9] found the year after, and Section 6 restores them. Of the family as a whole he adds [6]:

“Working constructions of type M with other rotations are not known yet.”

That sentence is the question this paper answers.

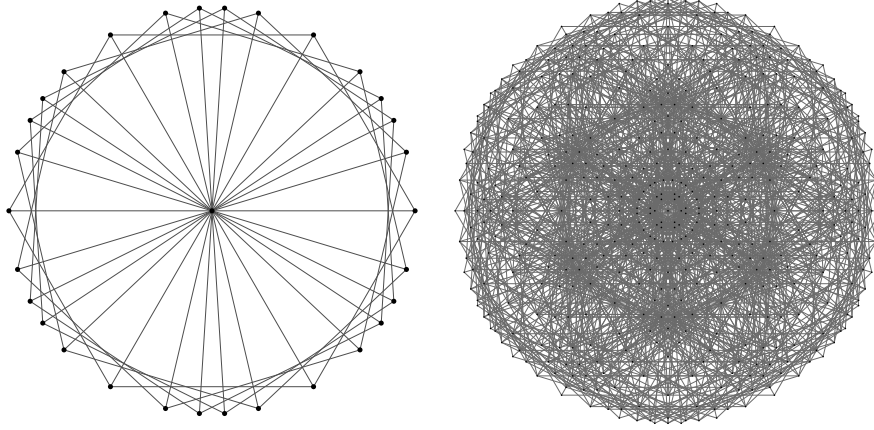


FIGURE 1. The Moser lattice R . Left: the origin, its 30 unit neighbours and every unit edge among those 31 points—six copies of a five-fan of half-width $\theta_3 = \arccos(5/6)$, and they are all the unit vectors R has—a finite check from (3), the equation $S = 144$ with $AB + CD = 0$ having exactly these thirty solutions in the lattice. Right: the ball of depth 2, 451 points and 1920 unit edges. R is dense in the plane—it contains $\frac{1}{3}\mathbb{Z} + \frac{\sqrt{33}}{3}\mathbb{Z}$ on the real axis and is closed under the rotation through 60° ; what is drawn is the unit-distance graph it carries, not the point set.

The rotations at issue. Polymath16’s algebraic formulation [7] indexes the natural rotations by $\omega_t = \exp(i \arccos(1 - 1/(2t)))$, the apex angle of a $\sqrt{t}, \sqrt{t}, 1$ isosceles triangle. The notation agrees with the one above rather than colliding with it: ω_1 and ω_3 in this sense have cosines $\frac{1}{2}$ and $\frac{5}{6}$, so they are exactly the two generators of R already named. We write θ_a for the same rotation at squared radius a , allowing any rational $a > 1/4$ —the square root is real for $a \geq 1/4$, and $a = 1/4$ is the degenerate endpoint where $\theta_a = -1$ and the shell pairing collapses, so we exclude it—and then $4 \mid a$ is to be read 2-adically, as $v_2(a) \geq 2$, which for integral a is ordinary divisibility:

$$\cos \theta_a = \frac{2a - 1}{2a}, \quad \sin \theta_a = \frac{\sqrt{4a - 1}}{2a}.$$

One symmetry, and the normalisation it forces. R is carried to itself by complex conjugation and by -1 , so for any $L, S \subseteq R$,

$$\overline{L \cup \theta_a(S)} = \overline{L} \cup \theta_{a'}(-\overline{S}), \quad a' = \frac{a}{4a - 1},$$

with $\overline{L}$ and $-\overline{S}$ again inside R . The identity behind it is $\theta_{a'} = -\overline{\theta_a}$, which one reads off $\cos \theta_{a'} = -\cos \theta_a$ and $\sin \theta_{a'} = \sin \theta_a$. Mirror images are isometric, so θ_a joins if and only if $\theta_{a'}$ does. The map $a \mapsto a/(4a - 1)$ is an

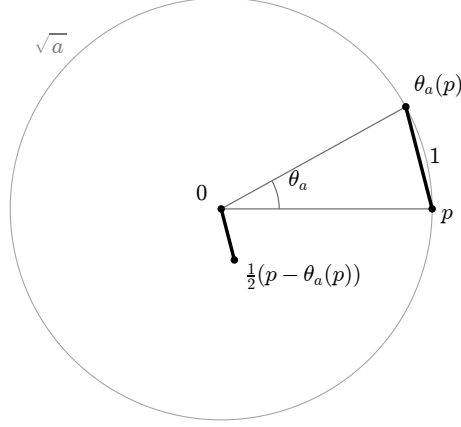


FIGURE 2. The spindle at squared radius a . Two shell points at distance $\sqrt{a}$ are carried onto each other by θ_a and land at distance exactly 1, which is what the rotation is for. Half that displacement has length $\frac{1}{2}$ always, since $|1 - \theta_a| = 1/\sqrt{a}$; whether it lies in the lattice is the whole content of Theorem 1. Drawn for $a = 4$, $\cos \theta_4 = 7/8$.

involution exchanging $(\frac{1}{4}, \frac{1}{2})$ with $(\frac{1}{2}, \infty)$ and fixing $\frac{1}{2}$, and it leaves $v_2(a)$ alone whenever the denominator of a is odd.

We normalise to $\cos \theta_a \geq 0$, that is to $a \geq \frac{1}{2}$, and every statement below about which θ_a join is a statement about such representatives. This is not cosmetic. It is the shell that breaks the symmetry: θ_4 spins a shell of 30 points, while its partner $\theta_{4/15}$ —which carries the mirror image of the very same 509-vertex graph—has an empty shell, since $15 \notin \{1, 3, 9\}$. Unnormalised, a criterion phrased through shells would admit one of the pair and reject the other.

Then $|p - \theta_a(p)| = |p||1 - \theta_a| = |p|/\sqrt{a}$, so a point and its own image are a unit pair exactly when $|p| = \sqrt{a}$: the rotation's own pairing lives on that *shell* and nowhere else, and Figure 2 is the whole mechanism in one picture. Other points do acquire cross edges—to images of *other* points, as Figure 5 shows—but if the shell is empty the rotation has nothing of its own to act on.

Which θ_a are useful has been approached experimentally. Voronov, Neopryatnaya and Dergachev [9] enumerate rotations carrying one subset of vertices onto another, and their first series—built on the Moser spindle, with the generators $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\frac{\sqrt{33}}{6} + \frac{\sqrt{3}}{6}i$, so on R —reports five that work (their Table 5), Heule's among them. So five working type-M rotations were known, not one.

They are, however, one orbit. All five have the form ρu with ρ Heule's rotation and u one of the 30 unit vectors of R : the arguments of the five u are $0, -\theta_3/2, -\theta_3, \theta_3/2 - 60^\circ$ and $\theta_3 - 60^\circ$, the last two matching ρu only up to complex conjugation. Multiplying by u does *not* fix R —four of the five move it, so the joint lattices genuinely differ—but of the five rotations exactly one has rational cosine, namely $\rho = \theta_4$ with $\cos = 7/8$. A spindle is precisely a rotation of rational cosine, so the other four are not spindles: in the family this paper classifies, exactly one working member was known. No criterion, and no classification, has been given for either family.

Results.

Theorem 1 (the divisibility law). *Let a be a positive rational and suppose the joint lattice $\Lambda_a = R + \theta_a(R)$ has rank 8—which by Proposition 3 is every a except those with $4a - 1 = 3k^2$ or $11k^2$ for a rational k . If $4 \nmid a$ then Λ_a admits a homomorphism 4-colouring. If $4 \mid a$ and the shell of R at radius $\sqrt{a}$ is non-empty, it admits none.*

The shell hypothesis carries only the second half, and it is not decoration: at $a = 8, 32, 40$ and 56 the shell is empty and a homomorphism colouring does exist, while at $a = 24$, empty as well, none does. An empty shell decides nothing either way, which is why Section 6 asks for occupancy separately.

A homomorphism 4-colouring properly colours the *entire* infinite lattice, at every radius, so:

Corollary 2. *If Λ_a has rank 8 and $4 \nmid a$, no subset of Λ_a is 5-chromatic; in particular no 5-chromatic $L \cup \theta_a(S)$ exists, at any size.*

Neither hypothesis is an obstacle to check case by case. The second is an explicit and recognisable exclusion, and the first is decided outright by Proposition 3: the rank is 8 unless $4a - 1$ is $3k^2$ or $11k^2$ for rational k , and those excluded a are not covered by anything here.

Proposition 3. *Let $V \subset \mathbb{C}$ be the $\mathbb{Q}$ -span of R . Then $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$, a field of degree 4 over $\mathbb{Q}$, and*

$$\theta_a \in V \iff 4a - 1 = 3k^2 \text{ or } 11k^2 \text{ for a rational } k.$$

If $\theta_a \in V$ then $\Lambda_a = R + \theta_a(R)$ has rank 4; otherwise it has rank exactly 8 and Theorem 1 applies. No intermediate rank occurs.

Proof. The chart (1) with rational A, B, C, D gives V the $\mathbb{Q}$ -basis $1, \sqrt{33}, i\sqrt{3}, i\sqrt{11}$. Since $i\sqrt{11} = (i\sqrt{3})\sqrt{33}/3$ and $(i\sqrt{3})^2 = -3$, this is $\mathbb{Q}(\sqrt{33})(i\sqrt{3})$, of degree 2 over $\mathbb{Q}(\sqrt{33})$ and 4 over $\mathbb{Q}$; in particular V is a field.

Now $\theta_a = c + is$ with $c = (2a - 1)/(2a) \in \mathbb{Q}$ and $s = \sqrt{4a - 1}/(2a)$, so $\theta_a \in V$ if and only if $is \in V$, that is $\sqrt{4a - 1} \in \mathbb{Q}\sqrt{3} + \mathbb{Q}\sqrt{11}$. Writing $\sqrt{4a - 1} = p\sqrt{3} + q\sqrt{11}$ and squaring gives $4a - 1 = 3p^2 + 11q^2 + 2pq\sqrt{33}$, so $pq = 0$ and $4a - 1$ is $3p^2$ or $11q^2$, with p, q rational. When a is an integer the rational k is forced to be one: if $4a - 1 = 3(u/v)^2$ in lowest terms then $v^2 \mid 3$, so $v = 1$; and then k is odd besides, since $4a - 1 \equiv 3 \pmod{4}$ while $3k^2$ and

$11k^2$ are 0 mod 4 for even k . For non-integer a the rational form is the one that bites: $a = 1/3, 7/3, 13/3, 31/3$ and $43/3$ all have $4a - 1 \in 3\mathbb{Q}^2$, all have rank 4, and all have a non-empty shell, so they are genuinely excluded from Theorem 1 and not vacuously.

If $\theta_a \in V$ then multiplication by it is a bijection of the field V , so $\theta_a V = V$ and $\Lambda_a \subset V$ has rank 4. If $\theta_a \notin V$, suppose $w \neq 0$ lay in $V \cap \theta_a V$, say $w = \theta_a v$ with $v \in V \setminus \{0\}$. Then $\theta_a = wv^{-1} \in V$, because V is a field—a contradiction. So $V \cap \theta_a V = 0$ and $\dim_{\mathbb{Q}}(V + \theta_a V) = 4 + 4 = 8$. $\square$

That R itself is not closed under multiplication— $\omega_3^2 = \frac{7}{18} + \frac{5\sqrt{11}}{18}i$ is not in it—costs nothing here: the argument needs only its $\mathbb{Q}$ -span to be. The excluded integers up to 100 are 1, 3, 7, 19, 25, 37, 61, 69 and 91, and the smallest excluded rationals with admissible denominator are $1/3, 7/3, 13/3, 31/3$ and $43/3$. A thin set either way, and not a surprising one, since θ_1 is ω_1 and θ_3 is ω_3 : these are rotations the lattice already contains. The direct rank computation agrees everywhere it has been run, on the integers and on the rational cases above; the accompanying code reports the range.

Figure 3 runs the law over the integer spindles to $a = 100$: the first row is what the lattice offers, the second what the law allows, and the third is the answer.

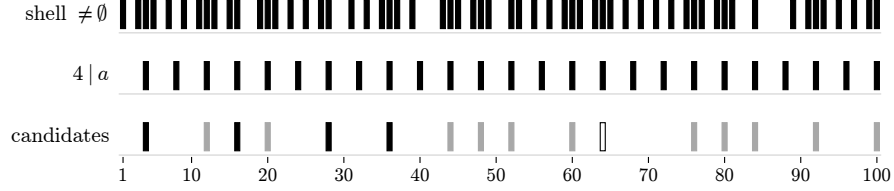


FIGURE 3. Theorem 1 as a sieve, over the integer spindles $a \leq 100$. Top: the 59 squared radii at which R actually has points, decided by the form (3) rather than by a ball, and in closed form by Theorem 16. Middle: the multiples of 4. Bottom: their intersection, the sixteen multiples of 4 with an *occupied* shell—the candidates, in the sense Section 6 fixes. The nine gaps there are the multiples of 4 whose shell is empty, $a = 8, 24, 32, 40, 56, 68, 72, 88, 96$; they lie outside this accounting altogether, because at an empty shell Theorem 1 is silent in both directions. The law is a kill, not a construction, so the bottom row also carries what became of each candidate: filled, the four given a verified 5-chromatic union in Section 6; grey, the eleven the stronger filter of Section 7 kills; open, $a = 64$, on which that filter degenerates at every modulus tried—it joins all the same, by Lemma 18, which builds a graph there rather than testing one.

Two words in the statement are used throughout in the following senses. A 4-colouring of R is *integral* if its colour classes are the cosets of a subgroup of index 4—equivalently, if it is a group homomorphism onto a group of order four, which is Gibbs’ construction and the only kind considered in this paper. It is *geometric* if whether a pair of points is monochromatic depends on the distance between them and nothing else; a geometric colouring is therefore carried to itself by every isometry of R fixing the origin, and in particular by multiplication by $\omega = e^{i\pi/3}$. Both conditions are used below, and used separately: integrality is what puts the question inside $R/2R$, and geometry is what forces the quotient to be Klein rather than cyclic.

Theorem 4 (the colourings are prime reductions). *The rational prime 2 splits in $\mathbb{Q}(\sqrt{33})$ as $-\pi_1\pi_2$ with $\pi_{1,2} = (5 \pm \sqrt{33})/2$. The lattice R is a module over $\mathcal{O} = \mathbb{Z}[(1+\sqrt{33})/2]$, and the two integral geometric 4-colourings of R —there are exactly two, as the proof below counts—are the reductions modulo π_1R and π_2R . In particular there are two of them because 2 splits into two primes.*

Theorem 4 is what makes the proof of Theorem 1 short: “this colour vanishes” becomes “this prime divides”. Section 6 exhibits six further joining spindles, so the answer to Parts’ question is that at least seven join in this family, four of them type M in his own sense. Two of the six are not new in the way the others are, and it is worth saying so before claiming them. Parts published two minimal monochromatic-pair gadgets, at distances $8/3$ and $8/\sqrt{3}$; doubling one about a terminal gives a 5-chromatic join at the rotation that spins it, so $\alpha_{64/9}$ and $\alpha_{64/3}$ already carry joins on 733 and 841 vertices. Chaining the first gadget with itself gives $\alpha_{256/9}$ the same way, as Section 6 sets out. All three of our rational spindles are therefore consequences of gadgets he had in hand, and what is added at them is the field they live in—which an earlier reading had wrongly ruled out—the ball-union form, and the arithmetic that decides in advance which rotations can work at all. For the three integer spindles θ_{16} , θ_{28} and θ_{36} we have found no prior construction, on the search Section 6 describes; θ_4 is the record and was known. Counting rotations rather than constructions, at least eleven join in the broader two-copy class: the five already known and these six. Eight of the eleven are type M in Parts’ sense, where he recorded one.

What is not new. The idea of colouring the lattice by a group homomorphism onto a four-element group is Philip Gibbs’, from Polymath16 in 2018 [7]: he showed the Moser ring admits one. Our criterion is that idea aimed at the *joint* lattice of a candidate rotation rather than at the Moser ring, and the contribution here is the classification it yields, not the tool. The ω_t notation and the algebraic framework are Polymath16’s.

That a lattice of this kind has exactly two integral geometric 4-colourings is Dúcz’s theorem [2], his Theorem 3.2, and the two colourings themselves are older still: Dúcz credits Hubai [5], who found them in the fourth Polymath16

thread in 2018 among 22 essentially different 4-colourings of the Moser ring. We claim neither. One caution about the attribution: Dúcz states his theorem for the lattice he defines, generated by $1, \omega_1, \omega_3$ and $\omega_1\omega_3$, which sits inside R with index 9—it misses $1/3$, and twelve of R 's thirty unit vectors—so his theorem and the count below are about different groups, and we do not transfer either to the other. The count below is proved here, for R , and stands on its own. What Section 4 adds is the *reason* for the number two— $33 \equiv 1 \pmod{8}$, so 2 splits into exactly two primes—which is a structural explanation neither of them gives: Dúcz's proof is a sublattice argument on coordinates mod 2 and contains no prime ideal and no number field.

Two more near misses worth naming, because a reader will wonder. De Grey's 2018 construction [1] does use a second rotation, by $2 \arcsin \frac{1}{8}$, which is our θ_{16} ; but it is applied about a point other than the origin—his graph L is K together with a copy of K rotated about a vertex A —so it is not a type-M join of two lattices, and the final graph there turns about $(-2, 0)$ by $\pi/2 \pm \arcsin \frac{1}{8}$. It is suggestive, not a second data point. And Parts [6] already writes his vertices over $\sqrt{33}$, in his own normalisation—denominator $4 \cdot 3^h$ where ours is 12—as $(A + B\sqrt{33} + iC\sqrt{3} + iD\sqrt{11})/(4 \cdot 3^h)$ subject to $A - B + C + D \in 4\mathbb{Z}$. (His coordinates are put in capitals here; lower-case a is the squared radius throughout this paper.) That congruence normalises coordinates inside one fixed lattice; the divisibility here is a condition on the *rotation*, deciding before any graph is built whether a join can be 5-chromatic at all. The ingredients were on the page; the test was not.

2. THE JOINT LATTICE, AND WHY NO BALL IS NEEDED

Every earlier test of a rotation built a ball of lattice points, found its unit edges and ran a SAT solver. That has a free parameter—the ball—and a negative answer about a ball is not an answer about the lattice: a rotation whose shell at radius $\sqrt{a}$ lies outside the ball is being dismissed on a graph that never contained a single pair $p, \theta_a(p)$. Section 6 exhibits six rotations that a ball at the record's own scale misses for exactly that reason.

No ball is needed. Write W for the 30 unit vectors of R . For a candidate θ , both halves of $L \cup \theta(S)$ lie in

$$\Lambda_\theta = \langle W \cup \theta(W) \rangle,$$

the group they and their images generate: a finitely generated subgroup of $(\mathbb{C}, +)$ with no radius in it. Its unit vectors are enumerable exactly: $q(z) = \text{Tr}(z\bar{z})$ is a positive-definite quadratic form on $\Lambda_\theta \otimes \mathbb{Q}$, and a unit vector has $z\bar{z} = 1$ and hence $q(z) = \text{Tr}(1)$, so the unit vectors lie *among* the integer points of one ellipsoid. That set is finite and can be listed exactly; sifting it by the exact condition $z\bar{z} = 1$ —which is two conditions where q is one—leaves the unit vectors, a complete list obtained without reference to any ball.

The criterion. Look for a group homomorphism $\varphi : \Lambda_\theta \rightarrow \mathcal{A}$ with $|\mathcal{A}| = 4$ and $\varphi(u) \neq 0$ for every unit vector u . If one exists then for any $p, q \in \Lambda_\theta$ at distance 1 the difference $p - q$ is a unit vector, so $\varphi(p) \neq \varphi(q)$ and $p \mapsto \varphi(p)$ is a proper colouring of the whole infinite lattice. Hence

$$\varphi \text{ exists} \implies \text{no subset of } \Lambda_\theta \text{ is 5-chromatic, ever.}$$

The converse fails—a lattice may be 4-colourable by a colouring that is not a homomorphism—so the test is a *one-sided kill*, and a survivor is a candidate and nothing more. There are two groups of order 4, the Klein group and $\mathbb{Z}/4$, and both must be ruled out.

A one-sided kill has one way to be wrong: killing something that works. Every rotation known to carry a 5-chromatic type-M graph is therefore a test of the criterion, and there are five—Heule’s and the four of [9]. Run against all five, the criterion returns *unknown* each time and kills none; the five joint lattices have rank 8 and carry 126, 108, 108, 90 and 90 unit vectors respectively. That is the only outcome consistent with their being 5-chromatic, and it is not automatic.

3. THE CRITERION IN CLOSED FORM

Proposition 5. *Suppose Λ_θ contains a vector of length $1/2$. Then no homomorphism onto a group of order 4—Klein or cyclic—avoids every unit vector.*

Proof. Let $|h| = 1/2$, so $2h$ is a unit vector of Λ_θ . In the Klein group every element satisfies $2x = 0$, so $\varphi(2h) = 2\varphi(h) = 0$ and the unit vector $2h$ is killed.

In $\mathbb{Z}/4$, $\varphi(2h) = 2\varphi(h) \neq 0$ forces $\varphi(h)$ odd, and that is already impossible. Write $\omega = e^{i\pi/3}$, a unit vector of R with $\omega R = R$, so $\omega\Lambda_\theta = \Lambda_\theta$ and $\omega h, \omega^2 h$ are half-unit vectors alongside h . Since $\omega^2 = \omega - 1$,

$$h + \omega h + \omega^2 h = 2\omega h \in 2\Lambda_\theta,$$

so $\varphi(h) + \varphi(\omega h) + \varphi(\omega^2 h) = 2\varphi(\omega h)$ is even, while three odd values sum to an odd one. $\square$

Both halves are unconditional, and the ω -identity is why: it is a relation of *odd* weight among half-unit vectors, and it exists wherever one half-unit vector does. Swept over every integer spindle to $a = 100$ that has one—sixteen of them—the relation is found every time, as it must be.

Remark 6. An earlier version of this section reported the relation as *failing* at $a = 48, 64$ and 80 , and called the cyclic half conditional on that account. The report was an artefact. The sweep collected the images of the half-unit vectors in $\Lambda_\theta/2\Lambda_\theta$ and discarded the zero class before testing; at those three values *every* half-unit vector lies in $2\Lambda_\theta$ —all 42, 30 and 60 of them—so discarding zero emptied the system, and the empty system read as no relation. A half-unit vector in $2\Lambda_\theta$ is in fact the strongest case of all: it is a relation of weight one, since no homomorphism sends 0 to an odd residue.

Half-unit vectors are therefore the object to count, and for a spindle they have a closed form. For p on the shell, p and $\theta_a(p)$ are both at distance $\sqrt{a}$, and

$$\frac{1}{2}(p - \theta_a(p))$$

has length exactly $1/2$ for every a , since $|1 - \theta_a| = 1/\sqrt{a}$. Now $p - \theta_a(p)$ always lies in Λ_a , being a difference of one element of R and one of $\theta_a(R)$; whether *half* of it does is another matter, and that is the whole content of the classification: $\frac{1}{2}(p - \theta_a(p)) \in \Lambda_a$ exactly when $p - \theta_a(p) \in 2\Lambda_a$.

4. THE ARITHMETIC OF THE MOSER LATTICE

4.1. A module over a real quadratic ring. In the chart

$$z = \frac{A + B\sqrt{33}}{12} + i \frac{C\sqrt{3} + D\sqrt{11}}{12}, \quad A, B, C, D \in \mathbb{Z}, \quad (1)$$

the lattice R is cut out by the congruences

$$(i) \quad A \equiv B \equiv C \equiv D \pmod{2}, \quad (ii) \quad A + B + C + D \equiv 2A \pmod{4}, \quad (2)$$

sixteen of the 256 residue classes mod 4. (The chart without them is a group containing R with index 16, so reading a chart count as a lattice count overcounts—at rational squared radii at least; Lemma 7 below says it never does at integer ones.) Squared lengths are

$$|z|^2 = \frac{(A^2 + 33B^2 + 3C^2 + 11D^2) + 2(AB + CD)\sqrt{33}}{144}, \quad (3)$$

so they lie in $\mathbb{Q}(\sqrt{33})$, and are *rational exactly when* $AB + CD = 0$, in which case they lie in $\frac{1}{144}\mathbb{Z}$. Write

$$S = A^2 + 33B^2 + 3C^2 + 11D^2$$

for the numerator, so a rational squared length is $S/144$. (This S is the form's value throughout Sections 4 and 5; the small half of a construction carries the same letter only outside them.)

At *integer* squared radii the congruences cost nothing, which is worth recording because it is what makes the chart safe to compute with there.

Lemma 7. *Let $A, B, C, D \in \mathbb{Z}$ satisfy $AB + CD = 0$ and $16 \mid S$. Then (2) holds automatically.*

Proof. Squares modulo 16 depend only on their bases modulo 8, so both hypotheses and both conclusions are determined by (A, B, C, D) modulo 8, and $8^4 = 4096$ classes decide the statement. Of those, 320 satisfy $16 \mid S$ while failing (2), and not one of the 320 has $AB + CD \equiv 0 \pmod{8}$. $\square$

If $|z|^2 = a$ is a positive integer then $S = 144a$, so $16 \mid S$ and $AB + CD = 0$, and Lemma 7 applies: every integral chart point at an integer squared radius already lies in R . Measured, the two agree exactly—identical counts at every $a \leq 300$, not merely the same emptiness.

The module structure is in fact a great deal more than a module structure.

Proposition 8. $3R$ is the maximal order $\mathcal{O}_V$ of $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$; that is, $R = \frac{1}{3}\mathcal{O}_V$. Writing $F = \mathbb{Q}(\sqrt{33})$ for the real quadratic subfield and $\bar{}$ for complex conjugation, which generates $\text{Gal}(V/F)$,

$$|z|^2 = z\bar{z} = N_{V/F}(z),$$

so a squared radius of R is one ninth of a relative norm of an algebraic integer.

Proof. Write coordinates over the $\mathbb{Q}$ -basis $(1, \sqrt{33}, i\sqrt{3}, i\sqrt{11})$ of V . Reducing the 30 generators of R to Hermite form and multiplying by 3 gives a $\mathbb{Z}$ -basis of $3R$:

$$b_1 = \frac{1}{4}(1, 1, 1, -1), \quad b_2 = \frac{1}{2}(0, 1, 0, -7), \quad b_3 = \frac{1}{2}(0, 0, 1, 1), \quad b_4 = (0, 0, 0, 1).$$

It contains 1, because $\frac{1}{3} \in R$ by (1) at $(4, 0, 0, 0)$, and it is closed under multiplication: the sixteen products $b_i b_j$ reduce to integer combinations of the b_k by the rules $\sqrt{3}\sqrt{11} = \sqrt{33}$, $\sqrt{3}\sqrt{33} = 3\sqrt{11}$ and $\sqrt{11}\sqrt{33} = 11\sqrt{3}$. So $3R$ is an order in V . (The same closure holds on all 900 products of unit vectors, where R itself fails: only 684 of them return.)

For the discriminant, $\text{Tr}_{V/\mathbb{Q}}(x) = 4x_1$ on these coordinates, since the four embeddings negate $\sqrt{33}$ and $i\sqrt{3}$ independently and kill every other basis vector. Hence

$$(\text{Tr}(b_i b_j)) = \begin{pmatrix} 5 & -22 & 4 & 11 \\ -22 & -506 & 77 & 154 \\ 4 & 77 & -14 & -22 \\ 11 & 154 & -22 & -44 \end{pmatrix}, \quad \det = 1089.$$

Now compare with the field. Put $F = \mathbb{Q}(\sqrt{33})$, so $V = F(i\sqrt{3})$ and $\text{disc}(F) = 33$ because $33 \equiv 1 \pmod{4}$. Two standard facts close the argument without computing $\text{disc}(V)$ separately. First, for any order $\mathcal{O}$ of V ,

$$\text{disc}(\mathcal{O}) = [\mathcal{O}_V : \mathcal{O}]^2 \text{disc}(V).$$

Second, the conductor–discriminant formula for the relative extension V/F of degree 2 gives $\text{disc}(V) = N_{F/\mathbb{Q}}(\mathfrak{d}_{V/F}) \text{disc}(F)^2$, so $\text{disc}(V)$ is a positive multiple of $33^2 = 1089$. Taking $\mathcal{O} = 3R$ and substituting,

$$1089 = [\mathcal{O}_V : 3R]^2 \cdot N_{F/\mathbb{Q}}(\mathfrak{d}_{V/F}) \cdot 1089,$$

which forces both factors to be 1. So $3R = \mathcal{O}_V$, and as a by-product $\mathfrak{d}_{V/F} = (1)$: V/F is unramified at every finite prime, which is what Theorem 16 will use.

The norm identity is the definition of $N_{V/F}$: complex conjugation fixes $\sqrt{33}$ and negates $i\sqrt{3}$, so it generates $\text{Gal}(V/F)$ and $z\bar{z} = N_{V/F}(z)$. $\square$

Two consequences follow, and one of them is the whole of Section 6’s criterion.

- R is a module over $\mathcal{O} = \mathbb{Z}[(1 + \sqrt{33})/2]$, the ring of integers of $\mathbb{Q}(\sqrt{33})$. This is a finite check and not a sample. The congruences (2) depend only on (A, B, C, D) modulo 4, and exactly 16 of the 256 classes satisfy them. Multiplication by $\sqrt{33}$ sends (A, B, C, D) to $(33B, A, 11D, 3C)$ —using $\sqrt{3}\sqrt{33} = 3\sqrt{11}$ and $\sqrt{11}\sqrt{33} = 11\sqrt{3}$ —and carries each of the 16 classes to one of the 16. For $(1 + \sqrt{33})/2$ the same map, halved, is integral because $A \equiv B$ and $C \equiv D \pmod{2}$ on those classes, and again preserves them. Since $\text{rank}_{\mathbb{Z}} R = 4$ and $\text{rank}_{\mathbb{Z}} \mathcal{O} = 2$, the lattice R has rank 2 over $\mathcal{O}$.
- R is not a lattice in the geometric sense. It is dense, for the reason given in Section 1; its short vectors are as short as one likes, and a depth-4 ball already contains one with $|v|^2 = (23 - 4\sqrt{33})/9 \approx 0.0024$. Any argument wanting a smallest vector must restrict to rational norms, where $\frac{1}{144}\mathbb{Z}$ supplies one.

The smallest rational squared norms of R , decided exactly from (3) together with (2), are

$$\frac{1}{9}, \frac{1}{3}, \frac{4}{9}, \frac{5}{9}, \frac{7}{9}, 1, \dots$$

so $1/9$ is the least, attained by six points, and it is the only one below $1/4$. The value $1/36$ is not attained at all, a fact used in Section 5.

4.2. The two colourings are prime reductions. Since $33 \equiv 1 \pmod{8}$, the prime 2 splits in $\mathbb{Q}(\sqrt{33})$:

$$2 = -\pi_1\pi_2, \quad \pi_1 = \frac{5 + \sqrt{33}}{2}, \quad \pi_2 = \frac{5 - \sqrt{33}}{2}, \quad N(\pi_i) = -2.$$

Because R is an $\mathcal{O}$ -module, $\pi_i R$ is a subgroup, and $[R : \pi_i R] = N(\pi_i)^2 = 4$, exactly the index a 4-colouring needs, where $[R : 2R] = 16$ is four times too big.

Proof of Theorem 4. First, the quotient has no choice but to be Klein. By integrality the colour classes are the cosets of a subgroup H of index 4. By geometry the isometry $\omega = e^{i\pi/3}$ of R permutes those classes, and it fixes the origin, so it fixes the class containing it: $\omega H = H$, and ω induces an automorphism T of R/H . Since $\omega^2 = \omega - 1$,

$$T^2 - T + I = 0.$$

If R/H were cyclic of order 4 then $T = \pm I$, the only automorphisms it has, and the left side would be I or $3I$ —neither zero in $\text{End}(\mathbb{Z}/4)$. So R/H is the Klein group, $2R \subseteq H$, and the search may be confined to subgroups containing $2R$.

That done, a homomorphism onto the Klein group killing no unit vector is the same thing as a subgroup of index 4 containing $2R$ and meeting none of the 30 unit vectors, so the whole question lives in $R/2R$ —and Proposition 8 says exactly what that quotient is. Multiplication by 3 carries R isomorphically onto $\mathcal{O}_V$ and $\pi_i R$ onto $\pi_i \mathcal{O}_V$, so $R/\pi_i R = \mathcal{O}_V/\pi_i \mathcal{O}_V$; and π_i

is inert in V , since the completion of F at π_i is $\mathbb{Q}_2$ —the prime 2 splits—and $-3 \equiv 5 \pmod{8}$ generates its unramified quadratic extension. Each quotient is therefore the field $\mathbb{F}_4$, and as π_1, π_2 are coprime the Chinese remainder theorem gives a ring isomorphism

$$R/2R \xrightarrow{\sim} R/\pi_1 R \times R/\pi_2 R = \mathbb{F}_4 \times \mathbb{F}_4,$$

under which the two prime reductions are the coordinate lines: $\pi_1 R/2R = 0 \times \mathbb{F}_4$ and $\pi_2 R/2R = \mathbb{F}_4 \times 0$.

Now the unit vectors. A unit vector u has $|u|^2 = 1$, so $v_{\pi_i}(u) = 0$ at both primes by Lemma 12: its class is invertible in each factor. There are $3 \cdot 3 = 9$ such classes, and all nine occur. The thirty unit vectors are the $\omega^k \varphi^j$ with $0 \leq k < 6$ and $|j| \leq 2$, for the same $\omega = e^{i\pi/3}$ and the half-angle $\varphi = e^{i\theta_3/2}$ of Section 1; modulo $2R$, $\omega^3 = -1 \equiv 1$ and $\varphi^{-1} \equiv \varphi^2$, so their classes are the nine with $k, j \in \{0, 1, 2\}$, and those nine are distinct. So the 9 is a ceiling attained, not a coincidence of a count.

Two lines finish the theorem. The image W of an index-4 subgroup containing $2R$ is an $\mathbb{F}_2$ -subspace of $\mathbb{F}_4 \times \mathbb{F}_4$ of dimension 2. If it meets no unit class then every element of W has a zero coordinate, so

$$W = (W \cap (\mathbb{F}_4 \times 0)) \cup (W \cap (0 \times \mathbb{F}_4)),$$

a union of two subgroups—and a group is never the union of two proper subgroups. Hence W lies inside one coordinate line, and both lines have $\mathbb{F}_2$ -dimension 2, so W is that line. Exactly two subgroups survive, and they are the prime reductions:

$$\ker \gamma_1 = \pi_1 R, \quad \ker \gamma_2 = \pi_2 R.$$

The count—two—is the one Dúcz reaches for his own lattice, by enumeration rather than by his sublattice argument, and reached here for R , which is not his lattice.

Both are geometric, which closes the circle: by Lemma 12, $v_{\pi_i}(|x|^2) = 2v_{\pi_i}(x)$ for $x \neq 0$, so $x \in \pi_i R$ if and only if $v_{\pi_i}(|x|^2) \geq 2$ —membership in either kernel is read off the squared length alone, and monochromatic pairs are decided by distance. $\square$

In coordinates, for checking: take the Hermite basis of R in the chart (1), rows in the order (A, B, C, D) ,

$$b_1 = (1, 1, 1, -1), \quad b_2 = (0, 2, 0, -14), \quad b_3 = (0, 0, 2, 2), \quad b_4 = (0, 0, 0, 4),$$

and write a class of $R/2R \cong \mathbb{F}_2^4$ as the integer whose bit i is the coefficient of b_{i+1} . Then

$$\pi_1 R/2R = \{0, 5, 9, 12\}, \quad \pi_2 R/2R = \{0, 3, 4, 7\},$$

the nine unit classes are the nine integers in neither list, and a sweep over all 35 two-dimensional subspaces of $\mathbb{F}_2^4$ —the enumeration the argument above replaces—returns those two and no others. (The masks depend on the basis and the bit order; both are fixed here so the numbers can be checked.)

Stability is the wrong invariant, and it is worth saying which one fails, because matching the $\mathcal{O}$ -stable subgroups against Dúcz's two is the natural first move and cannot work. Multiplication by $\sqrt{33}$ is the *identity* on $R/2R$, since $33 \equiv 1$, so it constrains nothing; multiplication by $(1 + \sqrt{33})/2$ leaves eleven of the thirty-five two-dimensional subspaces stable—the two eigenspaces and the nine sums of a line from each—not two. It is avoiding the unit vectors that cuts the field to two, and the ring structure above that names them.

The theorem counts the Klein colourings, and Section 8 needs the cyclic ones too, so count them here. A homomorphism $R \rightarrow \mathbb{Z}/4$ missing every unit vector is determined by its values on a basis; writing those as $\ell + 2s$ over $\mathbb{F}_2$ makes the condition one affine system per ℓ , a sweep over 2^4 rather than 4^4 , exactly as in Proposition 5's proof. The answer is small enough to print. In the Hermite basis $b_1, \dots, b_4$ fixed below, writing a homomorphism as the tuple $(\varphi(b_1), \dots, \varphi(b_4))$, exactly twelve of the 256 tuples kill no unit vector:

$$(0, 0, 2, 1), \quad (0, 1, 0, 0), \quad (1, 0, 1, 3), \quad (1, 1, 0, 0), \quad (1, 1, 2, 1), \quad (1, 3, 1, 3)$$

and their negatives. Up to negation—the only non-trivial automorphism of $\mathbb{Z}/4$ —there are therefore exactly *six*. So R carries eight homomorphism 4-colourings altogether, two Klein and six cyclic, each counted up to the automorphisms of its own group, and it is that 8 that Section 8 pairs with itself.

The colour condition is therefore divisibility in $\mathcal{O}$,

$$\gamma_i(x) = 0 \iff \pi_i \mid x, \tag{4}$$

proved above and confirmed directly, point by point and for both colourings, on a ball of the lattice; the accompanying code reports the size. The number of colourings is two *because 2 splits into exactly two primes*: the subgroups that avoid every unit class are the two prime reductions, and there are two of those because $33 \equiv 1 \pmod{8}$. That is a reason, and Dúcz's sublattice proof does not give one. Note also $\pi_1 R \cap \pi_2 R = \pi_1 \pi_2 R = 2R$: both colours vanish exactly on $2R$.

Figure 4 draws them.

5. PROOF OF THE DIVISIBILITY LAW

Assume $\Lambda = R + \theta_a(R)$ has rank 8, so $\Lambda = R \oplus \theta_a(R)$ and a homomorphism $\varphi : \Lambda \rightarrow \mathcal{A}$ is a pair (φ_1, φ_2) , one per summand, and each φ_i must avoid the 30 unit vectors of its own summand.

Take $\mathcal{A}$ to be the Klein group first. Its exponent is 2, so $2R \subseteq \ker \varphi_i$ automatically—no geometric hypothesis is needed and none is available, since a homomorphism avoiding the unit vectors need not be geometric: Section 4 exhibits six cyclic ones that are not. What is available is the enumeration inside Theorem 4: a subgroup of index 4 containing $2R$ and meeting none of the nine unit classes is $\pi_1 R$ or $\pi_2 R$. The image cannot

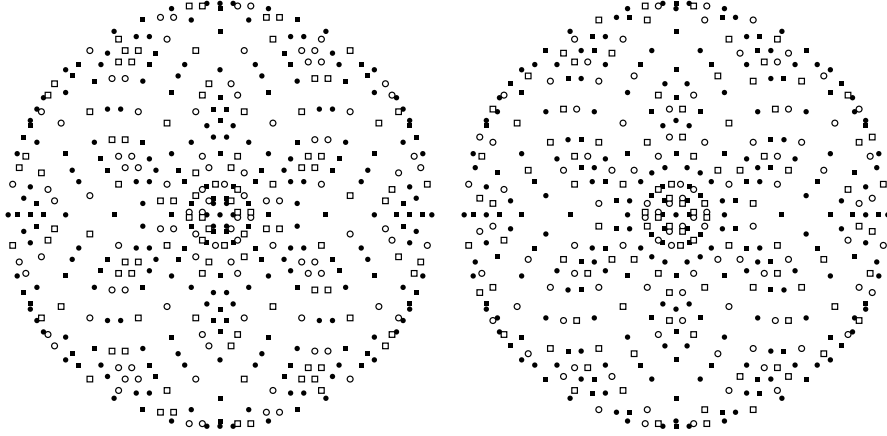


FIGURE 4. The two geometric 4-colourings of R , on the ball of depth 2, one glyph per colour. Each is proper—checked against the repository’s exact edge finder, zero monochromatic of the ball’s 1920 unit edges, which are left undrawn because at this density they hide the pattern—and by Theorem 4 they are the reductions modulo $\pi_1 R$ and $\pi_2 R$. The Galois conjugation $\sigma : \sqrt{33} \mapsto -\sqrt{33}$, which is what exchanges π_1 and π_2 , exchanges the two colourings as well: $\gamma_1(p) = \gamma_2(\sigma p)$ at every one of the 451 points, with no re-naming of colours needed. That is a statement about the lattice and not about the drawings— σ is no isometry, and moves 422 of the 451 points, changing the modulus at 312 of them—so the two panels are not congruent and should not be expected to look it.

be smaller than the whole Klein group either. Were it of order 2, then $1 + \omega + \omega^2 = 2\omega$ would give $\varphi_i(1) + \varphi_i(\omega) + \varphi_i(\omega^2) = \varphi_i(2\omega) = 0$, while each of the three terms is the one non-zero element of an order-2 group and their sum is that element again. So each φ_i is reduction modulo $\pi_1 R$ or $\pi_2 R$, and with the six automorphisms of the Klein group that is at most $2 \cdot 2 \cdot 6 = 24$ candidates, whatever a is. (Both reductions happen to be geometric, by Theorem 4, but that is a conclusion here and not a hypothesis.)

The cyclic target $\mathcal{A} = \mathbb{Z}/4$ is disposed of once and for all by Proposition 5, which Step 5 invokes; the six cyclic colourings counted in Section 4 are the ones that survive on R alone and play no part here. The question is which of them a *cross* unit vector $u = r + \theta(s)$, with $r, s \in R$ both non-zero, destroys. Figure 5 shows the two orbits these fall into.

Step 1: the two parts are parallel. Writing $w = r\bar{s}$,

$$|u|^2 = |r|^2 + |s|^2 + \frac{(2a - 1) \operatorname{Re} w + \sqrt{4a - 1} \operatorname{Im} w}{a}.$$

Separate the coefficients over $F = \mathbb{Q}(\sqrt{33})$. By (3) the squared norms $|r|^2$ and $|s|^2$ lie in F , and so does $\operatorname{Re} w$; the chart's imaginary part is $(C\sqrt{3} + D\sqrt{11})/12$ and $\sqrt{11} = \sqrt{33}/\sqrt{3}$, so $\operatorname{Im} w$ lies in $\sqrt{3}F$. Then $|u|^2 = 1$ reads

$$\alpha + \beta\sqrt{3(4a-1)} = 1, \quad \alpha, \beta \in F,$$

and rank 8 is exactly the statement that $\sqrt{3(4a-1)} \notin F$ —which by Proposition 3 fails only at $4a-1 = 3k^2$ or $11k^2$. Note this is the $\mathbb{Q}$ -span and not the field: at $a = 1/2$ the root $\sqrt{4a-1} = 1$ lies in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and the rank is 8 regardless. So $\beta = 0$, the $\sqrt{4a-1}$ term vanishes on its own: $\operatorname{Im} w = 0$, that is, $s = \lambda r$ for a real λ .

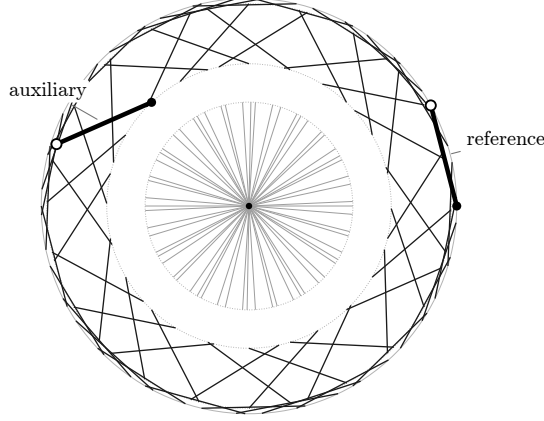


FIGURE 5. The entire interface of θ_4 : all 126 cross edges a depth-4 ball offers, computed from the coordinates. Sixty run to the shared origin—the inner star—thirty join two points of the shell at radius 2, and thirty-six run crosswise between the dotted radii $\frac{\sqrt{11}}{2} \mp \frac{\sqrt{3}}{6} = 1.3696$ and 1.9470 . Both halves here are the same ball. One edge of each non-trivial orbit is drawn heavy, its end in L filled and its end in $\theta_4(L)$ open. Every edge in the figure has length exactly 1.

Step 2: $\lambda = -1$ gives the shell. Then $u = (1 - \theta)r$, and $|1 - \theta| = 1/\sqrt{a}$ forces $|r|^2 = a$: the point r lies on the shell and u is a shell displacement. Since $s = -r$ and the colour group has exponent 2, $\gamma(s) = \gamma(r)$: the pair of colours is *diagonal*, (c, c) , and $\varphi(u) = \psi_1(c) + \psi_2(c)$, where ψ_1, ψ_2 are the two automorphisms—not the two primes, which keep the letter π throughout.

The colour group has exponent 2, so $\varphi(u) = 0$ exactly when $\psi_1(c) = \psi_2(c)$. At $c = 0$ that holds for *every* candidate, and $c = 0$ is exactly $r \in 2R$, both primes at once. For $c \neq 0$ it still holds for those candidates whose two automorphisms happen to agree at c , which is why the conclusion below asks for a relative 3-cycle and not merely for $c \neq 0$.

Step 3: $r \in 2R \iff 4 \mid a$. Here $S = 144a$ is an integer even when a is not, because A, B, C, D are. Then $v_2(S) = 4 + v_2(a)$, so $4 \mid a$ —that is $v_2(a) \geq 2$ —is exactly $64 \mid S$. Recall too that $|r|^2$ is rational exactly when $AB + CD = 0$. Three lemmas settle it.

Lemma 9. *For (A, B, C, D) satisfying (2) with $AB + CD = 0$ we have $16 \mid S$. Equivalently, every rational squared norm of R lies in $\frac{1}{9}\mathbb{Z}$.*

Proof. For odd x write $x^2 = 1 + 8k_x$; then $x \mapsto k_x \bmod 2$ is the homomorphism $\varepsilon : (\mathbb{Z}/8)^\times \rightarrow \mathbb{Z}/2$ with $\varepsilon = 0$ on ± 1 and 1 on ± 3 . If all four are odd then $S \equiv 8(\varepsilon(A) + \varepsilon(B) + \varepsilon(C) + \varepsilon(D)) \pmod{16}$, and $AB = -CD$ gives

$$\varepsilon(A) + \varepsilon(B) = \varepsilon(AB) = \varepsilon(-CD) = \varepsilon(C) + \varepsilon(D),$$

so the bracket is even. If all four are even then $S = 4S'$ with $S' = A'^2 + 33B'^2 + 3C'^2 + 11D'^2$, and mod 4, $S' \equiv [A'] + [B'] + 3[C'] + 3[D']$ with $[\cdot]$ the odd indicator; the congruences make the number of odd entries 0, 2 or 4, and of the six two-element patterns $\{A', B'\}$ and $\{C', D'\}$ are excluded by $A'B' + C'D' = 0$ (one product odd, the other even), while the rest give $1 + 3 \equiv 0$. $\square$

Lemma 10. *If A, B, C, D are all odd and $AB + CD = 0$ then $S \equiv 16 \pmod{32}$. The second hypothesis is not decorative: $(1, 1, 1, 3)$ gives $S = 136 \equiv 8$.*

Proof. Refine ε one step: for odd x write $x^2 = 1 + 8k_x$ as before and set $\eta(x) = k_x \bmod 4$, so $x^2 \equiv 1 + 8\eta(x) \pmod{32}$. From $AB = -CD$ we get $A^2B^2 = C^2D^2$, whose two sides are $1 + 8(\eta_A + \eta_B)$ and $1 + 8(\eta_C + \eta_D)$ modulo 32, the cross terms $64\eta\eta$ vanishing; so $\eta_A + \eta_B \equiv \eta_C + \eta_D \pmod{4}$. Expanding S modulo 32 gives $16 + 8(\eta_A + \eta_B - \eta_C - \eta_D) \equiv 16$. $\square$

Lemma 11. *If $4 \mid X^2 + 33Y^2 + 3Z^2 + 11W^2$ then $X + Y + Z + W$ is even.*

Proof. Modulo 4 a square is the odd indicator, so the form is congruent to $[X] + [Y] + 3[Z] + 3[W]$. One odd entry gives 1 or 3; three give $5 \equiv 1$ or $7 \equiv 3$. Neither is 0, so the number of odd entries is even. $\square$

Forward. If $r = 2w$ with $w \in R$ then $a = 4|w|^2$, and Lemma 9 gives $|w|^2 = T/9$ for an integer T , so $a = 4T/9$ and $v_2(a) = 2 + v_2(T) \geq 2$: that is $4 \mid a$, with no integrality of a used. (When a is an integer the same line reads $9 \mid T$.)

Backward. If $4 \mid a$ then $64 \mid S$, so $32 \mid S$, which Lemma 10 forbids for all-odd coordinates; by (2) they are all even, and with $A = 2A'$ and so on, $S = 4S'$ with $16 \mid S'$ and $A'B' + C'D' = 0$. The halves satisfy

(i): Lemma 11 gives an even number of odd entries among them, and two is impossible. Suppose exactly two of A', B', C', D' are odd. The pairs $\{A', B'\}$ and $\{C', D'\}$ are out at once: either would make one side of $A'B' = -C'D'$ odd and the other even. That leaves the four *mixed* pairs, one entry from $\{A', B'\}$ and one from $\{C', D'\}$, and they all behave the same way, for a reason visible in the coefficients of $S' = A'^2 + 33B'^2 + 3C'^2 + 11D'^2$:

$$1 + 3 = 4, \quad 1 + 11 = 12, \quad 33 + 3 = 36, \quad 33 + 11 = 44,$$

every one of them $\equiv 4 \pmod{8}$. An odd square is $\equiv 1 \pmod{8}$, so the two odd terms of S' contribute 4 modulo 8. For the two even ones, $A'B' = -C'D'$ forces them to have the same 2-adic valuation $k \geq 1$ —the odd entry on each side contributes nothing—so together they contribute $2^{2k}(\alpha b^2 + \beta d^2)$ with b, d odd and $\{\alpha, \beta\}$ the complementary pair of coefficients, which is again $\equiv 4 \pmod{8}$. At $k = 1$ that is $4 \cdot (8j + 4) \equiv 0 \pmod{16}$, and at $k \geq 2$ the factor 2^{2k} already kills it. Either way $S' \equiv 4 \pmod{8}$, so S' is 4 or 12 modulo 16 and never 0, against $16 \mid S'$. They satisfy (ii): when all odd, reducing $A'B' = -C'D' \pmod{4}$ with every entry in $\{1, 3\}$ leaves an odd number congruent to 3, which is exactly (ii); when all even it reduces to $A'' + B'' + C'' + D''$ being even, and $16 \mid S'$ gives $4 \mid S''$, so Lemma 11 supplies it. Hence $r \in 2R$. $\square$

Step 4: every other λ imposes nothing new. Both r and s lie in $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$, which Proposition 3 says is a field, so $\lambda = s/r$ lies in V too; being real it lies in $V \cap \mathbb{R} = \mathbb{Q}(\sqrt{33})$, the only real quadratic subfield. Write $\lambda = \alpha/\beta$ with α, β coprime in $\mathcal{O}$ —which is possible because $\mathcal{O}$ is a principal ideal domain, $\mathbb{Q}(\sqrt{33})$ having class number one—and put $v = r/\beta$. It lies in R : Bézout gives $x\alpha + y\beta = 1$, and then $v = xs + yr \in R$ since R is an $\mathcal{O}$ -module, with $\beta v = r$ and $\alpha v = s$. Substituting $r = \beta v$, $s = \alpha v$ into $|u|^2 = 1$ and clearing denominators,

$$|v|^2 [a(\alpha - \beta)^2 + (4a - 1)\alpha\beta] = a. \quad (5)$$

Lemma 12. *For non-zero $v \in R$ and π a prime of $\mathcal{O}$ above 2, $v_\pi(|v|^2) = 2v_\pi(v)$, where $v_\pi(v)$ is the largest k with $v \in \pi^k R$.*

Proof. First, $9|x|^2 \in \mathcal{O}$ for every $x \in R$ —Lemma 9 is its rational case—because $144|\cdot|^2$ is a quadratic form with values in $\mathcal{O}$, and both its values and its polarisations on the 30 unit vectors are divisible by 16 there: 30 checks and $\binom{30}{2} = 435$. So $v_\pi(144|x|^2) \geq 4 = v_\pi(144)$ throughout R .

Now write $v = \pi^k v_1$ with $\pi \nmid v_1$. Since π is real, $|v|^2 = \pi^{2k}|v_1|^2$, so it is enough that $|v_1|^2$ be a π -unit, that is that $v_\pi(144|v_1|^2) = 4$ exactly. Replacing v_1 by $v_1 + \pi w$ changes $144|v_1|^2$ by $144\pi(v_1\bar{w} + \bar{v}_1 w) + 144\pi^2|w|^2$, of valuation at least 5 by the previous paragraph, so a valuation of 4 depends only on the class of v_1 in $R/\pi R \cong \mathbb{F}_2^2$. Its three non-zero classes are represented by the six points of least norm, $|v|^2 = 1/9$ —namely $\pm \frac{1}{3}$ and $\pm \frac{1}{6} \pm \frac{i\sqrt{3}}{6}$ —and there $144|v|^2 = 16 = (\pi_1\pi_2)^4$, of valuation exactly 4. $\square$

Lemma 13. *Let π be a prime of $\mathcal{O}$ above 2, and let the denominator of a be odd—which Theorem 16 forces whenever the shell is non-empty, since it puts the denominator in $\{1, 3, 9\}$. If π divides none of v , α , β , then a is odd.*

Proof. The residue field $\mathcal{O}/\pi$ is $\mathbb{F}_2$, so $\alpha \equiv \beta \equiv 1$ there, hence $\alpha - \beta \equiv 0$ and $\alpha\beta \equiv 1$, and the bracket in (5) is $\equiv 4a - 1 \equiv 1 \pmod{\pi}$: writing $a = n/b$ with b odd makes $4n - b$ odd, so $v_\pi(4a - 1) = 0$. (With $4 \mid b$ that fails, which is why the hypothesis is stated.) So π does not divide it, and (5) gives $v_\pi(|v|^2) = v_\pi(a) = v_2(a)$. With $\pi \nmid v$, Lemma 12 makes the left side 0. $\square$

Corollary 14. *If $4 \nmid a$ then neither prime divides v . If moreover a is even then π_i divides exactly one of $r = \beta v$ and $s = \alpha v$, so by (4) exactly one of $\gamma_i(r), \gamma_i(s)$ vanishes and $\varphi(u) = 0 + c$ with $c \neq 0$: the vector imposes no condition on a candidate colouring both summands the same way—which is all Step 6 asks for—for every λ , rational or not.*

Proof. Write $a = n/b$ in lowest terms and clear b in (5):

$$|v|^2 [n(\alpha - \beta)^2 + (4n - b)\alpha\beta] = n,$$

a bracket that lies in $\mathcal{O}$ whatever the parities, where the bracket of (5) need not. By Lemma 12 the factor $|v|^2$ has valuation $2v_\pi(v)$, so the left side has valuation $2v_\pi(v) + v_\pi(\text{bracket})$; the bracket lies in $\mathcal{O}$, so that second term is at least 0, and comparing with the right gives $2v_\pi(v) \leq v_2(n)$ —an inequality, which is all the proof uses. If b is odd then $v_2(n) = v_2(a) \leq 1$ because $4 \nmid a$; if b is even then n is odd and $v_2(n) = 0$. Either way $v_\pi(v) = 0$.

Now let a be even. Lemma 13 puts π into one of v, α, β ; the first is excluded, and α, β are coprime, so exactly one of them is divisible—say $\pi \mid \beta$. Then $\pi \mid r$, while $\alpha \equiv 1$ and $v \not\equiv 0 \pmod{\pi}$ leave $\gamma_i(s) = \gamma_i(v) \neq 0$. $\square$

The first half of the corollary settles the whole of Steps 4 and 5 by itself, for every λ at once, and it is worth saying so before the case analysis that found it. Fix a prime π_i . Since $\mathcal{O}/\pi_i = \mathbb{F}_2$, an α the prime misses satisfies $\alpha \equiv 1$, so $\alpha v \equiv v$ modulo $\pi_i R$; hence $\gamma_i(s) = \gamma_i(\alpha v)$ is 0 if $\pi_i \mid \alpha$ and $\gamma_i(v)$ if not, and likewise $\gamma_i(r) = \gamma_i(\beta v)$. As α, β are coprime the prime misses at least one of them, leaving three cases and only three, for $4 \nmid a$:

- $\pi_i \mid \beta$. Then $\gamma_i(r) = 0$ while $\gamma_i(s) = \gamma_i(v) \neq 0$, so $\varphi(u) \neq 0$ for every candidate: the vector imposes nothing at all.
- $\pi_i \mid \alpha$. The same with r and s exchanged.
- Neither. Then $\gamma_i(r) = \gamma_i(s) = \gamma_i(v)$ and the pair is the diagonal (c, c) of Step 2, with $c \neq 0$.

So a cross vector with $\lambda \neq -1$ imposes nothing the shell displacements do not, whatever λ is—rational or not, a odd or even. That disposes of every λ , at odd and even a alike, and the proof can go straight to the conclusion. Appendix A records the finer analysis the trichotomy replaced, and the sweep that checks both against every rank-8 spindle to $a = 80$.

Step 5: conclusion.

Lemma 15. *For a shell point r , either both primes divide r —which happens exactly when $4 \mid a$ —or neither does.*

Proof. Both divide if and only if $r \in \pi_1 R \cap \pi_2 R = 2R$, which by Step 3 is $4 \mid a$. Suppose exactly one does, say $r = \pi_1 w$. Then $|w|^2 = a/\pi_1^2 = a(29-5\sqrt{33})/8$, so in the chart w has

$$S = 522a, \quad 2(AB + CD) = -90a.$$

A non-empty shell puts a in $\frac{1}{9}\mathbb{Z}$ by Lemma 9, and 9 divides both 522 and 90, so these are integers; it also makes $v_2(a) \geq 0$, so the two cases below exhaust $4 \nmid a$. If a is odd then $S \equiv 2 \pmod{4}$, and $S \equiv [A] + [B] + 3[C] + 3[D]$ takes the value 2 only on the patterns ooe and eeo, both of which violate (i). If $v_2(a) = 1$, write $a = 2a'$ with a' odd; then $S \equiv 4 \pmod{8}$, which all-odd coordinates cannot give since they force $S \equiv 0 \pmod{8}$, so all coordinates are even and $AB + CD = 4(A'B' + C'D')$, while $AB + CD = -45a = -90a'$ is $2 \pmod{4}$. $\square$

If $4 \mid a$ and the shell is non-empty—this is the only place the second hypothesis is used, and the only half of the theorem that needs it—Step 3 gives a shell point $r = 2w$, and then $(1 - \theta_a)w$ lies in Λ_a with length $\frac{1}{2}$, so Proposition 5 rules out both groups of order 4 at once. Concretely, the shell displacement $u = (1 - \theta_a)r$ has both colours zero, so $\varphi(u) = 0$ for all 24 Klein candidates and none survives. If $4 \nmid a$, Lemma 15 says no shell point is divisible by either prime, so both colourings give it a non-zero colour: the shell displacements impose only diagonal pairs (c, c) with $c \neq 0$, uniformly over the shell and for either choice of colouring. Fix one of the two and use it on both summands, which is what makes the pairs diagonal at all. By Step 4 nothing else constrains, and two automorphisms differing at every non-zero point satisfy $\psi_1(c) \neq \psi_2(c)$ —the relative permutation must be a 3-cycle, and two of the six are—so at least four of the twenty-four candidates survive, and a colouring exists. This proves Theorem 1. $\square$

6. SEVEN JOINING SPINDLES, THREE OF THEM NEW

The law leaves $4 \mid a$ with a non-empty shell as the candidates. Which a have one is decided by the form (3) and not by a ball, and has a closed answer, because R is $\frac{1}{3}\mathcal{O}_V$ and occupancy is a norm question.

Call a prime $p \neq 3, 11$ *obstructed* when it splits in $\mathbb{Q}(\sqrt{33})$ and stays inert in $\mathbb{Q}(\sqrt{-3})$. For odd p that is the pair of symbols

$$\left(\frac{33}{p}\right) = 1 \quad \text{and} \quad \left(\frac{-3}{p}\right) = -1,$$

and neither need be computed: since $33 \equiv 1 \pmod{4}$, quadratic reciprocity gives $\left(\frac{33}{p}\right) = \left(\frac{p}{33}\right)$, and $\left(\frac{-3}{p}\right) = 1$ exactly when $p \equiv 1 \pmod{3}$. Both are

therefore fixed by p modulo 33, and running the 20 classes coprime to 33 leaves five:

$$p \text{ is obstructed} \iff p \equiv 2, 8, 17, 29 \text{ or } 32 \pmod{33}.$$

The prime 2 is obstructed too, straight from the definition— $33 \equiv 1 \pmod{8}$ splits it, $-3 \equiv 5 \pmod{8}$ keeps it inert—and it lands in the class $2 \pmod{33}$ with the others, so the residue form covers it as well and nothing below makes it a separate case. Below 200 the obstructed primes are 2, 17, 29, 41, 83, 101, 107, 131, 149, 167, 173 and 197; the residue form is checked against the symbols for every odd prime below 20000.

Theorem 16. *Let $a = n/b$ be a positive rational in lowest terms. Then R has a point at squared radius a if and only if $b \in \{1, 3, 9\}$ and $v_p(n)$ is even at every obstructed p .*

Proof. By Proposition 8, a is a squared radius of R if and only if $9a = N_{V/F}(w)$ for some $w \in \mathcal{O}_V$. Write $\varepsilon = 23 - 4\sqrt{33}$, a generator of the totally positive units of F —the unit group is $\langle -1 \rangle \times \langle \varepsilon \rangle$ and -1 is negative at both real places, so those units are exactly the powers of ε . Three further facts are used, and it is worth saying where each comes from.

V/F is cyclic of degree 2 and unramified at every finite prime: that is the by-product of Proposition 8's discriminant comparison, not a new assumption.

$h(F) = 1$, in two lines. The Minkowski bound for $F = \mathbb{Q}(\sqrt{33})$ is $\frac{1}{2}\sqrt{33} < 2.88$, so every ideal class contains an integral ideal of norm at most 2, that is the unit ideal or a prime above 2. Both of those are principal, since $2 = -\pi_1\pi_2$ with $\pi_{1,2} = (5 \pm \sqrt{33})/2$ in $\mathcal{O}_F$. So the class group is trivial.

$\varepsilon = 23 - 4\sqrt{33}$ is the fundamental unit of F , and it is totally positive: $23^2 - 33 \cdot 4^2 = 1$, and both conjugates $23 \mp 4\sqrt{33}$ are positive. Since -1 is negative at both real places, the totally positive units are exactly the powers of ε .

$h(V) = 1$, by Minkowski again, and the computation is small enough to give in full. With $n = 4$, $r_2 = 2$ and $|\text{disc}(V)| = 1089$ the bound is

$$\left(\frac{4}{\pi}\right)^2 \frac{4!}{4^4} \sqrt{1089} = \frac{16}{\pi^2} \cdot \frac{3}{32} \cdot 33 < 5.02,$$

so every ideal class contains an integral ideal of norm at most 5. Which primes of $\mathcal{O}_V$ have norm that small? Above 2: each π_i is inert in V , so the prime above it has norm $N(\pi_i)^2 = 4$, and there is no prime of norm 2. Above 3: the prime $\mathfrak{p}_3$ of F ramifies, and splits in V into two primes of norm 3. Above 5: $33 \equiv 3 \pmod{5}$ is a non-residue, so 5 is inert in F and every prime above it has norm at least 25. Hence the only integral ideals of norm at most 5 are the unit ideal, the two primes of norm 3 and the two of norm 4, and it is enough to generate those four.

The ones above 2 are $\pi_1\mathcal{O}_V$ and $\pi_2\mathcal{O}_V$, principal already in $\mathcal{O}_F$. For those above 3, take

$$w = \frac{1}{2}(1 + i(2\sqrt{3} + \sqrt{11})) \in \mathcal{O}_V,$$

which lies in $3R$ at the chart point $(6, 0, 12, 6)$ and has

$$N_{V/F}(w) = |w|^2 = \frac{1}{4}(1 + (2\sqrt{3} + \sqrt{11})^2) = \frac{1}{4}(24 + 4\sqrt{33}) = 6 + \sqrt{33},$$

so $N_{V/\mathbb{Q}}(w) = 36 - 33 = 3$ and (w) is one of the two; its conjugate under $\sqrt{33} \mapsto -\sqrt{33}$ is the other. So the class group is trivial.

(PARI agrees, and `scripts/moser_is_ov.py` runs the check, but the argument above does not need it.)

Which primes of F are inert in V ? Exactly those above an obstructed p , together with the one above 11. That is the content of the two symbols: p splits in F and stays inert in $\mathbb{Q}(\sqrt{-3})$ precisely when the prime above it does not split in V .

Necessity. Let $9a = N(w)$ with $w \in \mathcal{O}_V$. Then $9a$ lies in $\mathcal{O}_F \cap \mathbb{Q} = \mathbb{Z}$, so $b \mid 9$. At a prime P of F inert in V , $v_P(N(w)) = 2v_{P'}(w)$ is even. For P above an obstructed p we have $v_P = v_p$ on rationals, so $v_p(9a)$ is even; as no obstructed p equals 3, $v_p(a)$ is even too. (Above 11 the prime ramifies in F , so $v_P = 2v_{11}$ is even for every rational and the condition is empty.)

Sufficiency. Suppose $b \mid 9$ and $v_p(a)$ is even at every obstructed p . Then $9a$ is a positive rational, hence totally positive in F , and has even valuation at every prime of F inert in V . At the two infinite places V is totally imaginary, so the local norms there are the positive reals, which $9a$ meets; at a finite prime that splits there is no condition, and at an inert one the condition is even valuation. By Hasse's norm theorem for the cyclic extension V/F , $9a = N(x)$ for some $x \in V^\times$.

To get an algebraic integer, build the ideal instead. Even valuation at the inert primes is exactly what lets $(9a)\mathcal{O}_F$ be written as $N_{V/F}(A)$ for an ideal A of $\mathcal{O}_V$; since $h(V) = 1$, $A = (w)$, and $N(w)$ generates $(9a)$, so $N(w) = 9a u$ for a unit u of $\mathcal{O}_F$. Both $9a$ and $N(w) = |w|^2$ are totally positive, so u is. The totally positive units of F are the powers of ε —the unit group is $\langle -1 \rangle \times \langle \varepsilon \rangle$ and -1 is negative at both places—and $\varepsilon = N_{V/F}(\eta)$ for

$$\eta = i(2\sqrt{3} - \sqrt{11}) \in \mathcal{O}_V,$$

which is purely imaginary, so $\bar{\eta} = -\eta$ and

$$N_{V/F}(\eta) = \eta\bar{\eta} = (2\sqrt{3} - \sqrt{11})^2 = 12 + 11 - 4\sqrt{33} = 23 - 4\sqrt{33} = \varepsilon,$$

while $N_{V/\mathbb{Q}}(\eta) = 23^2 - 16 \cdot 33 = 1$, so η is a unit. It lies in $\mathcal{O}_V = 3R$ because $\eta/3 = i(2\sqrt{3} - \sqrt{11})/3$ is the chart point $(0, 0, 8, -4)$. Replacing w by $w\eta^{-k}$ with $u = \varepsilon^k$ gives $N(w) = 9a$ exactly. $\square$

The ingredients are all small and all checkable: $h(F) = h(V) = 1$, $\text{disc}(V) = 1089$, the signature $(0, 2)$, and $N_{V/F}(\eta) = \varepsilon$, which the accompanying code verifies along with $3R$ being the maximal order. Where the parity comes from is set out below, after the density.

The denominator contributes nothing beyond being admissible, which is consistent: 3 is not obstructed, and the reason is local rather than a matter of ramification—11 is a square in $\mathbb{Q}_3$, so $\sqrt{-3}$ already lies in $\mathbb{Q}_3(\sqrt{33})$ and

the extension splits there, leaving no condition. At 11 the extension is inert, but 11 ramifies in $\mathbb{Q}(\sqrt{33})$, so a rational has even valuation at the prime above it and the condition is empty again. The restriction $b \in \{1, 3, 9\}$ is Lemma 9's corollary from Section 5, and with it the criterion covers *every* rational radius, not merely the integers. The three rational spindles found below are all $n = 64$ or 256 , powers of two with even exponent, and all three have non-empty shells.

Put beside Theorem 1 this describes the occupied-shell candidates off the factorisation of a , with no ball and no search. It does not say which candidates admit a 5-chromatic join: Section 7 kills eleven of the sixteen below 100; $a = 64$ is the one it cannot reach, and Lemma 18 settles that one by construction instead.

Corollary 17. *Let $a \geq 1/2$, normalised as in the introduction, and write $a = n/b$ in lowest terms. Then θ_a is a candidate—the law admits it and its shell is non-empty—if and only if*

$$b \in \{1, 3, 9\}, \quad v_p(n) \text{ is even at every obstructed } p, \quad v_2(n) \geq 2.$$

Only the last clause is new: 2 being obstructed, the middle clause already asks for v_2 even, and $4 \mid a$ adds $v_2 \geq 2$, so between them $v_2 \in \{2, 4, 6, \dots\}$. That single line accounts for the nine empty multiples of 4 below 100 without any computation: eight of them—8, 24, 32, 40, 56, 72, 88, 96—have v_2 odd, and the ninth is $68 = 2^2 \cdot 17$, where the obstructed prime 17 appears once. Checked against the form for every integer $a \leq 400$, with no exception.

It also settles how common candidates are, and the answer is contrary to appearances. Write $\mathcal{C}$ for the integer candidates. Each condition is a restriction at one prime and the conditions at distinct primes are independent, so for *any* finite set P of obstructed primes containing 2,

$$\bar{d}(\mathcal{C}) \leq \frac{1}{6} \prod_{\substack{p \in P \\ p \neq 2}} \frac{p}{p+1},$$

the factor $\frac{1}{6}$ being 2's own share— v_2 even and at least 2—which is why the product skips it. This is an upper bound for every P , not an asymptotic, and it is all that is needed. Five of the twenty classes mod 33 are obstructed, so those primes have Dirichlet density $\frac{1}{4}$ and $\sum 1/p$ diverges over them; letting P grow drives the product to zero, and the candidates have natural density 0—candidates meaning what Corollary 17 defines, spindles the law admits whose shell is occupied. Whether a rotation can join without a shell of its own is not settled here, so this is not yet a statement that almost no spindle joins.

The decay is slow, which is why the first hundred look generous at 16: measured, the density runs 0.160, 0.136, 0.1236, 0.1154, 0.1089 at $N = 10^2, \dots, 10^6$, against 0.10882 for the product truncated at $2 \cdot 10^6$. A Mertens estimate over the obstructed primes would give the rate $(\log N)^{-1/4}$, which the measurements fit; we do not prove it, and nothing below uses it. Density

zero is the claim, and it needs only the divergence—and only the *necessary* half of Theorem 16: for any finite set P of obstructed primes the occupied-shell candidates have upper density at most the local product over P , and letting P grow sends that to 0.

The local mechanism is worth seeing directly, since it is what Theorem 16’s proof turns into a norm condition. By Proposition 3 the $\mathbb{Q}$ -span of R is $V = \mathbb{Q}(\sqrt{33})(\sqrt{-3})$, and $|z|^2 = z\bar{z}$ is the norm of that quadratic extension, so a rational a is a squared radius in V exactly when it is a norm from V down to $\mathbb{Q}(\sqrt{33})$ —that is, when $a = x^2 + 3y^2$ is soluble there. An obstructed p splits in $\mathbb{Q}(\sqrt{33})$, so the local field is $\mathbb{Q}_p$ itself and both local valuations are v_p ; and -3 is a non-square there, so the local extension is unramified quadratic. Its norm group is the elements of even valuation, which is where the parity comes from. At an *inert* p there is no condition at all, whatever -3 does: the local field is then already the unramified quadratic extension of $\mathbb{Q}_p$, so it contains $\sqrt{-3}$ and the norm map is onto. That is why the definition asks for both symbols and not just the second. The passage from local solubility to a norm of an algebraic *integer* is where $h(V) = 1$ and the unit η enter, and that is the content of the proof above; Lemma 7 is what makes the chart and R literally the same points at integer radii, so no index-16 correction intervenes when one computes with coordinates. Independently of the proof, the criterion was checked against the form for every $a \leq 400$ and for targeted larger values including $17^2, 17^3, 29^2, 41^2, 101^2$ and the powers of 2 up to 2^8 , with no exception.

Of the integer spindles to $a = 100$ this leaves the shells at $a = 8, 24, 32, 40, 56, 68, 72, 88, 96$ empty, and the rest are candidates. The rational ones are constrained before any search: by Lemma 9 every rational squared norm of R lies in $\frac{1}{9}\mathbb{Z}$, so a rational shell radius a has denominator 1, 3 or 9 and nothing else. Every normalised rational spindle with any other denominator spins an *empty* shell, in one line and with no computation—the denominator-5 family among them. That is not a death: Theorem 1 is silent at an empty shell, as the remark after it says. What it does mean is that such a rotation has no pair $p, \theta_a(p)$ of its own to work with, so nothing here supplies it a construction either. Without the normalisation the sentence would be false: $\theta_{4/15}$ has an empty shell and carries the mirror of the record.

Seven of the candidates yield working constructions. The first is the record and was known; of the other six, all three rational ones follow from Parts’ two published monochromatic-pair gadgets, as the chaining below shows, and so carry joins without any of this paper’s machinery. For the three integer spindles θ_{16} , θ_{28} and θ_{36} we have found no prior construction: we searched [6], [9], [4], [3] and [2] and their reference lists, the Polymath16 threads including [5] and [8], and the arXiv listings in combinatorics and metric geometry, to September 2026. That is a statement about our search, not a proof of priority—this literature is partly distributed across blog comments and data files, and we would be glad to be corrected. Each union

below is verified *not* 4-colourable from the coordinates, with exact arithmetic and a fresh solver—that is $\chi \geq 5$. We do not claim $\chi = 5$ for the full unions: that needs an upper bound as well, which we have only where a proper 5-colouring or vertex-criticality has been checked. They are not all type M in Parts’ sense: he asks for joining edges of two or more kinds, and counting the joining unit vectors of Λ_a by the radii of their two ends gives 2, 4, 3 and 2 kinds for θ_4 , θ_{16} , θ_{28} and θ_{36} , but a single kind—the shell pairs and nothing else—for all three rational spindles. So four are type M as he defines it, three of them new, and the other three are joins of two copies whose whole interface is the shell. The same count run on the five rotations already known returns 2 for each of them, on 66, 48, 48, 30 and 30 joining edges—their joint lattices have 126, 108, 108, 90 and 90 unit vectors, of which 60 run to the shared origin in every case and are excluded here, as they are for the spindles. They are type M as well, and since θ_4 is one of them the two lists together give $4 + 5 - 1 = 8$ type-M rotations where Parts recorded one. Three of the seven spindles have fractional a , and are written α_a instead of θ_a purely as a reminder of that: they are the same rotations of Section 1, with $\cos = (2a - 1)/(2a)$ and $\sin = \sqrt{4a - 1}/(2a)$, at $a = 64/3$, $64/9$ and $256/9$. The seven are:

rotation	cos	field	depth	radius	vertices
θ_4	7/8	$\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$	—	—	509, minimised
θ_{16}	31/32	$\mathbb{Q}(\sqrt{3}, \sqrt{7}, \sqrt{11})$	4	4.0000	29113
θ_{28}	55/56	$\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{37})$	6	5.3113	156577
θ_{36}	71/72	$\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{13})$	6	6.0000	175825
$\alpha_{64/3}$	125/128	$\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{23})$	7	4.6383	187225
$\alpha_{64/9}$	119/128	$\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$	3	2.6817	5809
$\alpha_{256/9}$	503/512	$\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{1015})$	6	5.3506	158257

The last column is not one measurement, and it is worth saying how the six were built, because no vertex-selection or minimisation search entered the full-ball constructions reported here: the halves are whole balls, not chosen subsets of one. The two parameters were found by taking the smallest depth whose ball reaches the shell at $\sqrt{a}$ —a ball stopping short of it holds no pair $p, \theta_a(p)$ and cannot work, so there is nothing to choose below that—and then the radius is the ball’s own. For each rotation, take the points of R reachable as a sum of at most d unit vectors and lying within radius r , and glue that set to its own θ_a -image at the origin: L and S are the *same* ball, so the union has $2|L| - 1$ vertices. The two parameters are the depth and radius columns above, and they are not quoted from a build log—`scripts/ball_params.py` recovers them from the shipped coordinates, by splitting off the half that lies in R and finding the smallest d whose ball at that radius is exactly that half. The radius is the half’s own, so it is the shell’s $\sqrt{a}$ where the shell is the outermost thing in the ball (θ_{16} , θ_{36}) and a little beyond where it is not. No vertices were selected, dropped or tuned; what is reported is that the whole ball works, not that a good subset of it was found. Parts’ 509

is a different kind of object—a minimised graph, its two halves pared down to 374 and 136—though it shares the origin the same way, which is why all seven counts are odd. The six here have not been minimised except where Section 8 says otherwise (θ_{16} to 2901, $\alpha_{64/9}$ to 1408). The six compare with each other and none of them with 509.

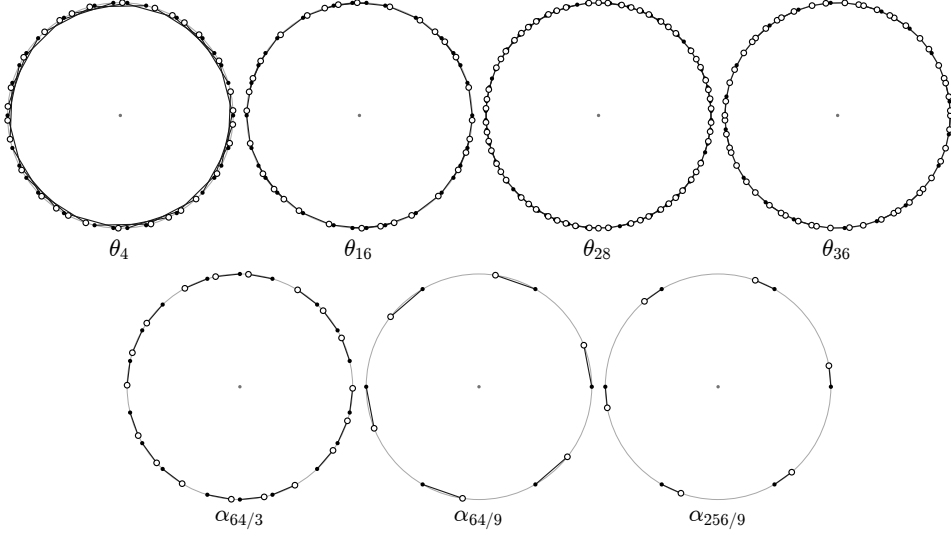


FIGURE 6. The seven joining rotations, each on its shell at radius $\sqrt{a}$: the lattice points at squared radius a (filled), their images under the rotation (open), and the unit edge every pair acquires. The shells hold 30, 30, 60, 54, 18, 6 and 6 points. Panels are drawn to a common circle size, so what varies across them is the shell and the rotation angle, not the scale; since $|1 - \theta_a| = 1/\sqrt{a}$ the chord is 1 in every panel and the angle closes as a grows.

Figure 6 shows each of the seven acting on its own shell. Here $\alpha_{64/9}$ is the smallest union by a wide margin (Figure 7), and its shell sits at distance $8/3$. That is not a free number. Parts' inventory holds exactly two minimal monochromatic-pair gadgets, 367 vertices at distance $8/3$ and 421 at $8/\sqrt{3}$, and the three rational spindles here spin *both*: $\alpha_{64/9}$ the first, $\alpha_{64/3}$ the second, with $\alpha_{256/9}$ at $16/3$, twice the first. They do not scatter, and the reason is that a gadget chains.

Lemma 18 (chaining a monochromatic pair). *Let $M \subset R$ be finite with two marked points 0 and $t \neq 0$ such that every proper 4-colouring of M gives them the same colour, and let $n \geq 1$ be such that $a = n^2|t|^2$ is rational and exceeds $\frac{1}{4}$, so that θ_a is one of the rotations of Section 1. Put $M_n = \bigcup_{j=0}^{n-1} (M + jt)$. Then $M_n \subset R$, every proper 4-colouring of M_n gives 0 and nt the same colour, and $|M_n| \leq n(|M| - 1) + 1$. Consequently, writing*

$$a = n^2|t|^2,$$

$$M_n \cup \theta_a(M_n)$$

is not 4-colourable, on at most $2n(|M| - 1) + 1$ vertices.

Proof. R is a group and $t \in R$, so every translate $M + jt$ lies in R and so does the union; $a > \frac{1}{4}$ makes $\sqrt{4a-1}$ real, so θ_a exists, and $a \in \mathbb{Q}$ makes it a spindle. In a proper 4-colouring the copy $M + jt$ forces jt and $(j+1)t$ to agree, so 0 and nt agree; the copies overlap in at least the shared marked point, which gives the count. The same argument inside $\theta_a(M_n)$ forces $\theta_a(0) = 0$ and $\theta_a(nt)$ to agree, and the two copies share the origin, so nt and $\theta_a(nt)$ both carry the colour of the origin. But $|nt|^2 = a$ puts nt on the shell, so $|nt - \theta_a(nt)| = 1$ and they are adjacent. $\square$

Parts' 367-vertex gadget is such an M , with $t = 8/3$: every one of its vertices lies in R , and forcing its two terminals apart leaves it not 4-colourable, which is his monochromatic-pair property re-proved here rather than cited. So $n = 1, 2, 3$ give joins at $a = 64/9, 256/9$ and 64, on at most 733, 1465 and 2197 vertices—the first two are the rational spindles above, reached without any of this paper's machinery, and *the third is* $a = 64$, the one integer spindle Section 7 cannot decide. It is decided: it joins. The 421-vertex gadget at $8/\sqrt{3}$ chains the same way and gives $\alpha_{64/3}$ at $n = 1$.

What these constructions deliver is $\chi \geq 5$, which is not the same as $\chi = 5$; the following disposes of the difference once, for all of them.

Lemma 19. *Let $G = L \cup \theta(S)$ be a finite non-4-colourable join sharing the origin. Then some induced subgraph of G containing the origin is a join of the same form and has chromatic number exactly 5.*

Proof. Among the induced subgraphs of G that contain the origin and are not 4-colourable—a non-empty family, since G is one—take one, H , with fewest vertices. Then $\chi(H) \geq 5$. For the other direction pick any $v \neq 0$ of H ; by minimality $H - v$ has a proper 4-colouring, and giving v a fifth colour extends it to H , so $\chi(H) \leq 5$. And H is a join of the same form: putting

$$L_H = L \cap V(H), \quad S_H = \theta^{-1}(V(H) \cap \theta(S)),$$

we have $V(H) = L_H \cup \theta(S_H)$ with $L_H \subseteq L$ and $S_H \subseteq S$, and $0 \in V(H)$ by construction. $\square$

So every entry of the table gives a 5-chromatic join, of at most the size listed; whether the *full* union has chromatic number 5 rather than more is a separate question, settled here only where a proper 5-colouring or vertex-criticality has been checked.

None of this improves the record—733 and 2197 are far above 509—and none of it is a search. It is the reason the filters of Section 7 were never going to settle $a = 64$: the construction was already available from a gadget published in 2020, and no colouring argument can exclude what exists.

A search confined to a pool has a reach, and that reach is the constraint: a ball must *reach* radius $\sqrt{a}$ or it holds no pair $p, \theta_a(p)$ at all, and all seven

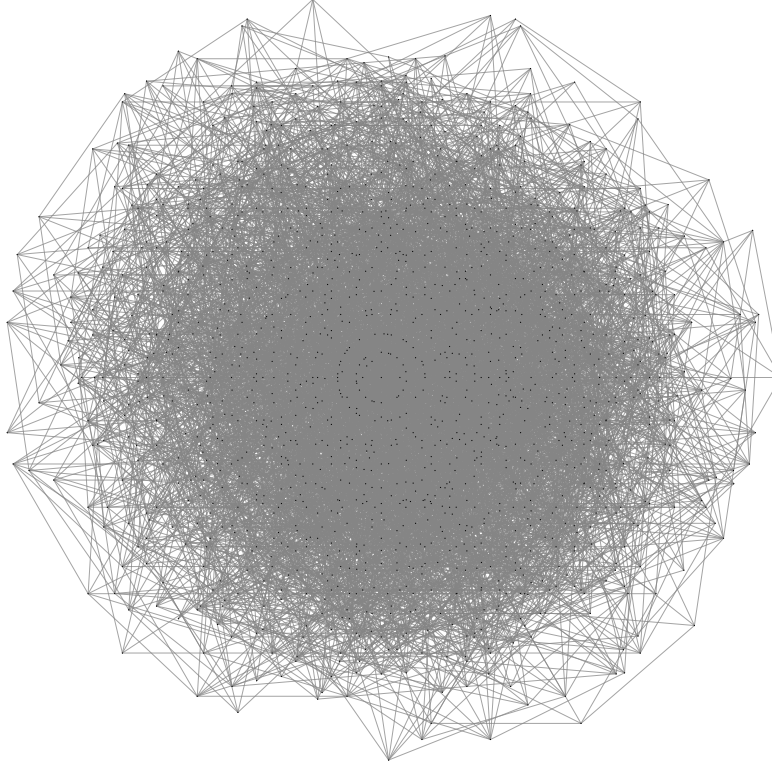


FIGURE 7. $\alpha_{64/9}$ after minimisation: 1408 vertices and 7013 edges, drawn from its coordinates. It is 5-chromatic and vertex-critical—asked of each of the 1408 in turn, no vertex can be deleted without a proper 4-colouring appearing—and it is the smallest we reach by minimising a ball union, against 2901 for θ_{16} . It is *not* the smallest known at this rotation: doubling Parts’ 367-vertex monochromatic-pair gadget about one terminal gives a 5-chromatic graph on 733, and an earlier version of this caption claimed otherwise. Nor is 1408 a record of any kind—it is 2.77 times Parts’ 509, which is a θ_4 graph.

constructions above use such pairs—for the three rational spindles they are the *only* joining edges there are, as the count of kinds showed. So a ball stopping short could not have produced these graphs. (It does not follow that no construction at those rotations avoids the shell; Theorem 23 says every *minimum* interface uses a shell pair, but only for θ_4 .) The scale at issue is small, and can be read off the record itself rather than guessed: Parts’ 509 vertices lie inside radius 2.5244, and 488 of them inside radius 2—the other 21 sit at the four radii 2.0238, 2.2909, 2.432 and 2.5243. A

search built to the record's own reach therefore holds θ_4 's shell, at radius 2, and no other. The next is $\alpha_{64/9}$'s at 2.67, and θ_{28} 's is at 5.29, which at this lattice's working density is a ball of 78289 points. We make no claim about what radius any particular published sweep used. Figure 8 sets the seven shells against the record's reach.

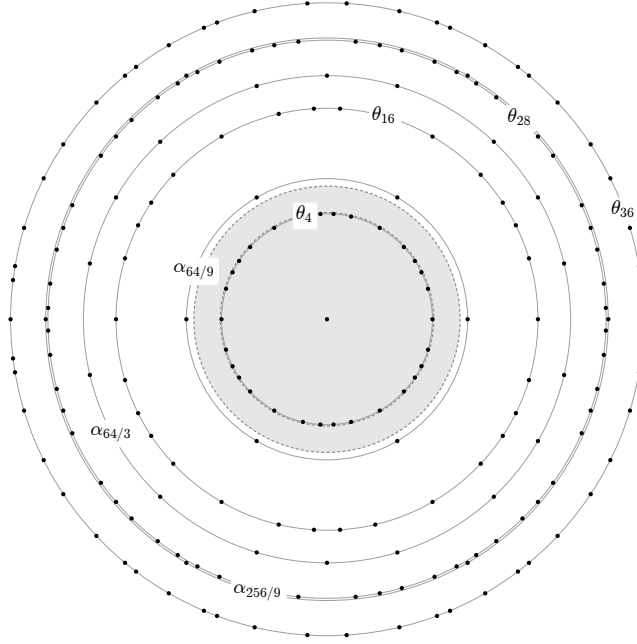


FIGURE 8. Six of the seven shells lie outside the record's own reach. The same shells at their true relative radii; the shaded disc is the reach of the record graph itself, radius 2.5244, and the dashed circle at 2 is where 488 of its 509 vertices stop. Only θ_4 's shell, at radius 2, lies within the disc. A point and its own image are a unit pair only at radius $\sqrt{a}$, so a ball stopping short of that radius contains no such pair—and every construction in Section 6 uses one—however carefully the search enumerates angles.

This does not make the record easier, and interface richness predicts nothing in either direction. Count a rotation's *new* unit vectors as those of Λ_a that R did not already have, so θ_4 's 126 become 96; of those, 30 are $\theta_4(R)$'s own and 66 genuinely join the two copies. Upwards: θ_{28} creates 150 against that 96, and its smallest known union has 156577 vertices. Downwards: $\alpha_{64/9}$ creates the fewest of the seven, 36, of which only six are genuinely joining—exactly the six its shell forces, with no second orbit at all—and its union is 5-chromatic all the same, and the smallest of the six built here at 5809. Nor does poverty predict smallness: $\alpha_{64/3}$ creates 48 and its union

runs to 187225. Richness says a rotation has material, not that a small graph follows, nor that a poor rotation fails.

7. A STRONGER FILTER, AND WHAT SURVIVES IT

The criterion of Section 2 uses group homomorphisms, which are exactly the *linear* proper colourings of R/mR . Every proper colouring of that finite quotient pulls back to the whole lattice, linear or not, so using all of them is strictly stronger, and it still needs no ball. A periodic colouring gives a cross pair (p, q) its colours through $p \bmod mR$ and $q \bmod mR$ alone, so all it can see is which *residue pairs* occur; and (α, β) occurs exactly when the coset $(\alpha - \theta\beta) + m\Lambda$ contains a unit vector of the joint lattice.

One degeneracy must be excluded: if the pair $(0, 0)$ occurs—equivalently if Λ holds a vector of length $1/m$ —then both halves give the centre’s coset the same colour and the test cannot kill. At $m = 2$ that condition is precisely the half-unit criterion of Section 3, so everything Proposition 5 disposes of is degenerate at $m = 2$; $m = 4$ and $m = 8$ are where it discriminates.

Applied to the integer spindles to $a = 100$, it kills θ_a for $a = 12, 20, 44, 48, 52, 60, 76, 80, 84, 92, 100$. The modulus is not the same for all of them, and saying which matters, because a rotation degenerate at one modulus is not thereby a survivor. Nine die at $m = 4$, over $4^4 = 256$ residue classes. The remaining two, $a = 48$ and $a = 80$, are degenerate at $m = 4$ and at $m = 6$ —the pair $(0, 0)$ occurs at both—and die at $m = 8$, over 4096 classes. Each kill is recorded as a witness naming its modulus: the two colourings and every occurring residue pair, re-checked to be proper, to agree at the shared origin, and to disagree on all 69 pairs. That leaves $\theta_4, \theta_{16}, \theta_{28}, \theta_{36}$ — and θ_{64} , which is left for the other reason: the pair $(0, 0)$ occurs at every modulus we tried, 4, 6, 8 and 12, so the filter never gets to speak. It never could have: Lemma 18 exhibits a 5-chromatic join at $a = 64$, and no colouring argument excludes what exists. That was forced rather than unlucky. $R \cap \mathbb{Q} = \frac{1}{3}\mathbb{Z}$, and the vector $(8/m)(1 - \theta_{64})$ has length $(8/m)|1 - \theta_{64}| = 1/m$ and lies in Λ_{64} as soon as $8/m \in \frac{1}{3}\mathbb{Z}$, which holds for every $m \mid 24$. (It has to be the unit vector 1 and not an arbitrary one: R is not a ring, and $\frac{1}{3}\omega_3$ is outside it.) All four moduli divide 24. Moduli with $m \nmid 24$ —5, 7, 9, 10, 16—escape this particular degeneracy, though nothing says another does not await there. The cases $a = 12, 20, 44$ were already known 4-colourable by ball tests, which is the check on the method; the other eight had never been tested, needing balls of radius 6.9 to 10. Each kill is a certificate, and a finite one: the two periodic colourings are checked against *every* residue pair that occurs, which by the criterion above is every coset of $m\Lambda$ holding a unit vector of the joint lattice. That is the whole infinite union, not a sample of it. As corroboration the same pair is then run against the exact edge set of a ball holding the whole shell—for θ_{12} , zero monochromatic edges of 867054; for θ_{20} , zero of 2076192—which tests the implementation rather than the claim.

So the divisibility law is necessary but far from sufficient. For the sixteen occupied-shell *integer* spindles that Theorem 1 admits to $a = 100$ —the candidates as Section 6 defines them—the account is now complete: eleven die to a test that uses every periodic colouring, four are the constructions of Section 6, and $a = 64$ joins by Lemma 18. The empty-shell multiples of 4 lie outside that tally, since the law is silent at an empty shell, and they are worth settling separately rather than leaving implicit. There are nine of them below 100, and both filters have now been run on all nine. Six die: $a = 8, 32, 40, 56, 68$ and 96 each carry a homomorphism 4-colouring—in fact 96 of them—so no subset of the joint lattice is 5-chromatic there either. Three survive both: $a = 24, 72$ and 88 admit no homomorphism 4-colouring at all, and the periodic filter proves at modulus 4 that no pair of 4-periodic colourings kills them, over the 66 residue pairs that occur in each. Those three are genuine candidates, on the same footing as the survivors above and with no construction known; they are not counted in the sixteen, because the law never admitted them.

That the filters can speak here at all is worth a sentence. An empty shell means no point is paired with its own image, so the spindle mechanism is absent—but Λ_a still has rank 8 and 72 unit vectors, and the cross edges they give pair *distinct* points. An earlier version of the accompanying code declined to run the periodic filter at an empty shell, on the ground that the question did not arise. It does arise, and these three are what the refusal was hiding. Nothing here covers the rational candidates—three of which work—or the finer lattices, or rotations outside the spindle family, where Section 10 reports both tools blind.

8. WHAT EVERY CONSTRUCTION MUST CONTAIN

The law constrains the rotation. The same eight colourings constrain the *construction*, and in a way that does not depend on which rotation is used.

Let $G = L \cup \theta(S)$ with L and $\theta(S)$ meeting only at the origin, so the only edges between the halves are the cross edges of Figure 5. Both halves sit in the same lattice R , so both are coloured by the eight homomorphisms counted in Section 4—two Klein, six cyclic—whatever θ is. Pair any colouring of L with any colouring of S , permute the four colours on the second, and the result properly 4-colours G *unless* some cross edge comes out monochromatic. A 5-chromatic G must therefore carry a set of cross edges meeting the monochromatic set of every admissible pairing: a minimum hitting set, decided exactly.

Which pairings are admissible matters. The halves share the origin and a homomorphism sends it to colour 0 on both sides, so only the 6 permutations fixing 0 give a well-defined colouring of G ; the other 18 colour the shared point twice over. Counting all 24 imposes a constraint no colouring realises and inflates the bound.

No ball is needed for any of this, which is worth establishing before the count, because it is what makes the bound a statement about the lattice.

Proposition 20. *Suppose $\Lambda_a = R \oplus \theta_a(R)$ has rank 8. Then $(p, \theta_a(q)) \mapsto p - \theta_a(q)$ is a bijection from the cross edges of $R \cup \theta_a(R)$ —pairs $p \in R$ and $\theta_a(q) \in \theta_a(R)$ at distance 1 from each other—onto the unit vectors of Λ_a . There are finitely many, though the union is dense. Moreover such an edge is monochromatic under a homomorphism φ of Λ_a exactly when $\varphi(u) = 0$, so what a cross edge contributes depends only on u .*

Proof. If $|p - \theta_a(q)| = 1$ then $u = p - \theta_a(q)$ is a unit vector of Λ_a . Conversely the direct sum writes a unit vector uniquely as $u = r + \theta_a(s)$, and then $p = r$, $q = -s$ is the only pair with $p - \theta_a(q) = u$. Writing $\varphi = (\varphi_1, \varphi_2)$ on the two summands, $\varphi(u) = \varphi_1(p) - \varphi_2(\theta_a(q))$, which vanishes exactly when the two ends receive the same colour. $\square$

The unit vectors of Λ_a lie among the integer points of one ellipsoid and are enumerated exactly from it, so the entire infinite union has a *finite and complete* cross-edge list. For θ_4 it has 126 entries, of which 60 meet the shared origin and 66 do not, and the $8^2 \cdot 6 = 384$ admissible pairings leave 235 distinct monochromatic patterns, none empty. Measured against balls of 5677, 8131, 10369 and 14323 points, the cross-edge count is 126 every time, as Proposition 20 requires.

Theorem 21. *No 5-chromatic $L \cup \theta_4(S)$ has four or fewer cross edges, for any $L, S \subseteq R$ whatever: the hitting-set instance is unsatisfiable at 4 and satisfiable at 5.*

Proof of Theorems 21–23. These three are computations, and what has to be computed is fixed by Proposition 20 and the count of colourings above, with no ball and no choices left. The objects are: the 126 unit vectors of Λ_4 , of which 60 meet the shared origin and 66 do not; the 8 homomorphism 4-colourings of R , two Klein and six cyclic; the $8^2 \cdot 6 = 384$ admissible pairings of one colouring per half with a colour permutation fixing 0; and the 235 distinct non-empty monochromatic patterns those pairings cut out of the 66. Theorem 21 is then the statement that the minimum set meeting all 235 has size 5, Theorem 22 that there are 1920 such sets, and Theorem 23 the profile and factorisation of those 1920.

Each step is finite and exact. `scripts/interface_bound_exact.py` 4 enumerates the unit vectors from the trace form, builds the 235 patterns and runs the minimum hitting set, reporting unsatisfiability at 4 and a solution at 5; `scripts/interface_matchings.py` enumerates all optimal sets, checks each is a perfect matching, and computes the profile and the $30 \times 8 \times 8$ factorisation. Both print every intermediate count named above, so a reader can check them one at a time rather than trusting a single verdict. What the three theorems do *not* say is that some 5-chromatic graph realises five cross edges: the 1920 defeat every pairing of homomorphism colourings, which is necessary and not known to be sufficient. $\square$

Theorem 22. *There are exactly 1920 such five-edge sets. None meets the shared origin—an edge out of it is never monochromatic, since a homomorphism sends the origin to 0 and no unit vector to 0, so the 60 cross edges at the centre appear in no pattern at all. Every one of the 1920 is a perfect matching, five edges on ten vertices; that is inherited rather than earned, because the remaining 66 cross edges are themselves a perfect matching.*

Theorem 23. *In the taxonomy of Figure 5 all 1920 have one profile: exactly one reference edge, both ends at radius 2, and exactly four auxiliary edges at radii $\frac{\sqrt{11}}{2} \mp \frac{\sqrt{3}}{6}$, two in each crosswise direction. So the radius of all ten interface vertices is fixed before any search begins. The count factors as*

$$1920 = 30 \times 8 \times 8 :$$

any of the 30 reference edges, and then, for that reference edge, one of exactly 8 admissible auxiliary pairs in each direction, every one of the 64 combinations occurring. The 8 depends on the reference edge—across all 30 there are 72 pairs per direction—so the three choices are not independent, only the last two. Of the $\binom{66}{5} = 8936928$ five-subsets of the usable cross edges, these 1920 are 0.021%.

The bound is one-sided in the safe direction: the 235 patterns come from homomorphism colourings only, and any colouring of the halves that is not one would add patterns and could only raise the bound. Five is a floor, not an estimate. And by Proposition 20 it is a floor over the whole lattice and not merely over a pool: there is no larger ball to try, since 126 is all the cross edges $R \cup \theta_4(R)$ has.

Two things it does not do. It does not rank rotations by promise— θ_{16} has the weakest obstruction measurable here and its smallest known graph has 2901 vertices, while θ_{28} ties θ_4 at five and its smallest has 156577 against 509. Both are bounds over the lattice: Proposition 20 makes the whole computation ball-free, since the pattern a cross edge contributes depends only on its unit vector, and run that way θ_4 , θ_{16} , θ_{28} and θ_{36} offer 66, 66, 120 and 90 usable cross edges with bounds 5, 1, 5 and 5. And it does not point at the record. Measured on Parts' 509-vertex graph itself, which is $L \cup \theta_4(S)$ with $|L| = 374$, $|S| = 136$ and exactly 18 cross edges, all 384 pairings are non-empty—the strongest check the method gets, since an empty one would be a proper 4-colouring of a 5-chromatic graph—and *six* of those 18 edges already defeat every closed-form colouring, themselves a perfect matching. Five is not reachable inside the record's own interface: the 1920 five-edge matchings live among the ball's 126 cross edges and the record carries only 18 of them, so five is a floor over the lattice and not a prediction about any particular graph. The other twelve are there to defeat colourings that are not homomorphisms, which is exactly what this argument cannot see.

9. THE SAME QUESTION ON ANOTHER LATTICE

Nothing in Section 2 is about the Moser lattice. The criterion needs a finitely generated subgroup of $(\mathbb{C}, +)$, its unit vectors, and a rotation, so the question can be asked wherever those exist. Haugland's Moser-spindle-free graph [3] is the case that matters at the moment: de Grey has raised it publicly as a possible route past 509, and asked in particular about using more than two of its sets of 42 vectors [8].

Which lattice, exactly. The construction lives in $\mathbb{Q}(\zeta_{420})$, of degree 96. Write L_1 for the subgroup of $(\mathbb{C}, +)$ generated by the 42 vectors $\zeta_{420}^{10j} = \zeta_{42}^j$ —all the 42nd roots of unity—and, for $k > 1$,

$$L_k = \left\langle \rho^s \zeta_{42}^j : 0 \leq s < k, 0 \leq j < 42 \right\rangle, \quad \rho = \frac{-\sqrt{3} + i(4 \cos \frac{\pi}{7} - 1)}{4 \sin \frac{2\pi}{7}} \zeta_3^{-1},$$

the multiplier that carries one set of 42 to the next. The formula is read off Haugland's own parametrisation [3], in which the second set of 42 is the first rotated by the argument of $(-\sqrt{3}, 4 \cos \frac{\pi}{7} - 1)$; that vector has length exactly $4 \sin \frac{2\pi}{7}$, which is what makes the displayed quotient an algebraic number of modulus exactly 1, and the factor ζ_3^{-1} aligns his indexing of the 42 directions with ζ_{42}^j . It reproduces the angle he reports, 0.42363201413287, to fourteen places. So ρ lies in $\mathbb{Q}(\zeta_{420})$, has modulus exactly 1, and is not a root of unity: its argument is 3.6311° . Which of these lattices de Grey's question and Haugland's own graph concern is a matter for [8] and [3]; what is defined here is L_k , and every claim below is about one of them. *Everything reported below about individual rotations—the table, the survival of θ_2 , the unit and half-unit counts—is computed in L_1 , and none of it is claimed for L_k with $k > 1$ or for whichever of these Haugland's own graph uses; the reader should take his paper for that. What is computed for several k is only the rank and the covolume, and it is labelled where it appears.*

Degree 96 looks prohibitive and is not: the $\mathbb{Q}$ -span of L_1 is $\mathbb{Q}(\zeta_{42})$, of degree $\varphi(42) = 12$. More sets do not raise it— L_1 to L_4 all have rank 12—and what they change is the covolume, which falls by a factor of 7 each time, so de Grey's suggestion is not cosmetic. A twelve-dimensional Fincke–Pohst enumeration is routine, and the criterion ports.

The span being a field again, Proposition 3's argument transfers verbatim, and settles the rank here too.

Proposition 24. *$L_1 + \theta_N(L_1)$ has rank 12 if $\theta_N \in \mathbb{Q}(\zeta_{42})$ and rank 24 otherwise, with no intermediate value; and*

$$\theta_N \in \mathbb{Q}(\zeta_{42}) \iff 4N - 1 = 3k^2 \text{ or } 7k^2.$$

Proof. The rank dichotomy is the proof of Proposition 3 with $\mathbb{Q}(\zeta_{42})$ in place of V : a non-zero element of the intersection would express θ_N as a quotient of two field elements. For the criterion, $\theta_N = c + is$ with c rational, so $\theta_N \in \mathbb{Q}(\zeta_{42})$ iff $\sqrt{-(4N - 1)} \in \mathbb{Q}(\zeta_{42})$. Since $42 = 2 \cdot 21$ with 21 odd, $\mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$, whose Galois group $(\mathbb{Z}/21)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/6$ has exactly three

subgroups of index 2; the three quadratic subfields are therefore $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{21})$. A square root of a negative number rules out the real one, leaving $-(4N - 1) = -3k^2$ or $-7k^2$. $\square$

Measured against the direct rank computation: $N = 1, 2, 7, 16$ give rank 12 and $N = 4, 9$ give rank 24, exactly as the criterion says. It also explains θ_4 , Haugland's own rotation: $4 \cdot 4 - 1 = 15$ is neither $3k^2$ nor $7k^2$, so its join is a genuine rank-24 object and not a symmetry.

Which spindles exist there at all is settled first, and arithmetically: θ_N needs $\sqrt{4N - 1}$ in the field, and $\mathbb{Q}(\zeta_{420})$ contains $\sqrt{m}$ exactly for the eight squarefree divisors m of 105. Of $N \leq 16$ that leaves $N = 1, 2, 4, 7, 9, 16$; every other N is excluded before any computation.

Then the criterion has a blind spot, and it is worth stating in general because it is not special to this lattice.

Proposition 25. *Let Λ be any subgroup of $(\mathbb{C}, +)$ closed under $\omega = e^{i\pi/3}$ and containing a unit vector g , and let $a = 4m^2$ for a positive integer m . Then $\Lambda + \theta_a(\Lambda)$ contains a vector of length $\frac{1}{2}$, so by Proposition 5 the criterion cannot kill θ_a : neither group of order 4 admits a homomorphism avoiding the unit vectors. Both lattices here are ω -closed— R by construction, and Haugland's because ω is one of its 42 roots of unity.*

Proof. $|2mg| = 2m = \sqrt{a}$, so $2mg$ lies on the shell, and

$$h_g = \frac{1}{2}(2mg - \theta_a(2mg)) = m(1 - \theta_a)g$$

lies in $\Lambda + \theta_a(\Lambda)$ with length $m|1 - \theta_a| = m/\sqrt{a} = \frac{1}{2}$. Proposition 5 does the rest, both halves of it, since Λ is ω -closed. $\square$

Haugland joins by θ_4 , whose shell radius is 2, so Proposition 25 applies and the criterion says nothing there—not a failed computation but a proof that this particular test is silent. Measured and consistent: the joint lattice $L_1 + \theta_4(L_1)$ has rank 24 and 210 unit vectors, of which exactly 42 lie in $2(L_1 + \theta_4(L_1))$, one per generator, as the proposition predicts.

The useful consequence is that the test has content precisely at the rotations Proposition 25 misses, and the classification above supplies them. In L_1 :

N	shell	$\sqrt{4N - 1}$	units	half-unit vectors	verdict
1	42	$\sqrt{3}$	42	0	dead, by a Klein colouring
2	84	$\sqrt{7}$	84	0	survives
4	—	$\sqrt{15}$	210	42	survives by Proposition 25

So θ_2 is the one non-trivial survivor found: it has no half-unit vector, so it gets no free pass, and a complete certificate search—affordable at rank 12, where this lattice's own rank-24 joins are not—returns nothing. And θ_2 is a rotation nobody has been able to use, because it needs $\sqrt{7}$, which the classical field $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$ does not contain. Here it is available and it passes a test with content.

Of the remaining three, $N = 16$ needs no computation at all: its shell radius is 4, so $a = 4m^2$ with $m = 2$ and Proposition 25 gives it the same free pass as θ_4 . That leaves $N = 7$ and $N = 9$, and we stop there honestly—but we can now say what stops us, which is not what it looked like. It is not the dimension. By Proposition 24, $4 \cdot 7 - 1 = 27 = 3 \cdot 3^2$, so θ_7 stays at rank 12, the same as the cases that finish in a second. What differs is the denominator: the Fincke–Pohst target is $96N \text{den}^2$, and the joint lattice’s denominator is 1 at $N = 1$, 2 at $N = 2$ and 7 at $N = 7$, so θ_7 searches a target 343 times θ_1 ’s where θ_2 ’s is 8 times it. It did not finish in ninety minutes. ($N = 9$ is rank 24 as well, so it is harder on both counts.) Reaching either needs the shell and the unit vectors described arithmetically rather than enumerated—the cyclotomic analogue of the chart (1), which exists but which we have not written. Nor is any of this a construction: θ_2 survives, and surviving is being a candidate.

10. WHAT IS NOT CLAIMED

The criterion is a one-sided kill. A survivor is a candidate; only an explicit 5-chromatic union makes it a construction, which is why Section 6 exhibits graphs rather than citing survivals.

The law is about *this* lattice, and Section 9 is the evidence: θ_2 dies on the Moser lattice and survives on L_1 , the lattice generated by the 42nd roots of unity. The divisibility law does not transfer, and no attempt is made to transfer it.

The rank-8 hypothesis excludes rotations that map R ’s span into itself— θ_3 is ω_3 , θ_1 is ω_1 —where the cross decomposition does not arise. Proposition 3 says exactly which those are, $4a - 1 = 3k^2$ or $11k^2$, so the hypothesis is recognisable at sight rather than checkable per rotation, and it is proved rather than assumed.

Proposition 5 is unconditional for both groups of order 4. An earlier version of it was not, and the remark in Section 3 records why: the three exceptions it reported were a bug in the sweep, not a feature of the lattice.

Theorem 16 is proved, through Proposition 8: R is $\frac{1}{3}\mathcal{O}_V$, so occupancy is a question about relative norms and Hasse’s theorem answers it. An earlier version of this paper stated it as verified and not proved, having missed that $3R$ is the maximal order. Theorem 1 does not—it is proved—and the constructions of Section 6 are exhibited rather than predicted, so nothing in Section 6 rests on the criterion either.

The choice of R is a choice. Taking $|j| \leq J$ instead of $|j| \leq 2$ in the generating family gives, for each J , a rank-4 lattice, and these nest: $30 \subset 42 \subset 54 \subset \dots$. The index over ours is 9^{J-2} , checked for $J \leq 5$, and the unit vectors are exactly the $6(2J + 1)$ generators, checked for $J \leq 3$. Neither is proved here for all J ; counting generators does not by itself rule out a further unit vector, and nothing below depends on the general statement. Heule’s 553-vertex large part lives in the 42-vector member. Everything here

is about the 30-vector one, which is what the record and every construction in Section 6 use; the law is not claimed for the finer members, and the criterion would have to be recomputed on each.

Nothing here reaches the rotations of [9] that are not spindles, and we checked rather than assumed it. Their four are ρu for a unit vector u of R ; the whole family $\{\rho u\}$ is 30 rotations, of which six—the sixth roots of unity—fix R and so give back θ_4 's own joint lattice, and exactly two are spindles, ρ and $-\rho$. Every one of the 30 has a rank-8 joint lattice with 126, 108 or 90 unit vectors. Run against all 30, the criterion of Section 2 kills *none*, and the stronger periodic filter of Section 7 kills none either, degenerating at $m = 4$ and $m = 8$ alike. So both tools are blind outside the spindle family: against the integer spindles the law removes three a in four and the periodic filter eleven of the sixteen survivors, and against $\{\rho u\}$ they remove nothing at all. Whatever decides that family is not in this paper.

No claim is made about the record. The bound 509 is unbeaten, and nothing here brings it closer. What is settled is an obstruction—no spindle of rank 8 with $4 \nmid a$ can be used at all, the rank-4 exceptions of Proposition 3 being untouched—together with a proved criterion for occupancy, eleven further exclusions and seven constructions. What is not settled is the classification those would have to add up to: rotations outside the spindle family are untouched.

APPENDIX A. THE ODD CASE, AND THE SWEEP

The trichotomy at the end of Step 4 settles the odd case together with the even one, for every λ at once. This appendix records the finer analysis it replaced—which says not merely that each $\lambda \neq -1$ is harmless but which λ occur—and the computational sweep that checks both against every rank-8 spindle to $a = 80$. Nothing in the paper depends on either.

Which λ occur at odd a . For odd a Lemma 13 says nothing, and two sub-cases exhaust what occurs.

(a) *λ rational.* Write $\lambda = \alpha/\beta$ with α, β coprime integers, $\beta \geq 1$. A rational λ makes $m = |v|^2$ rational, so m is a rational squared norm of a non-zero point of R , hence $m \geq 1/9$ by Section 4. Rearranged, (5) reads $m[a(\alpha + \beta)^2 - \alpha\beta] = a$. Both signs end up bounding the same quantity, $|\alpha| - |\beta|$, but for different reasons.

Negative λ , $\lambda = -p/q$ with $p, q \geq 1$, $d = q - p$. Here (5) reads $ad^2 + pq = a/m$. With $pq \geq 1$,

$$d^2 < 1/m \leq 9, \quad \text{so} \quad |d| \leq 2,$$

and $d = 0$ is $\lambda = -1$; $|d| = 3$ is impossible, not merely unobserved. If $|d| = 1$ then exactly one of p, q is even, so 2 and hence both primes divide one of r, s , while the other is an odd multiple of v and Corollary 14 keeps its colour non-zero: $\varphi(u) = 0 + c$ with $c \neq 0$ once more. (Here a is odd, so it is the first half of the corollary that applies, not the second.) If $|d| = 2$

then p and q are coprime and differ by 2, so both are odd; $4a + pq = a/m$ forces $m < 1/4$, and $1/9$ is the only squared norm below $1/4$ (Section 4), so $m = 1/9$ exactly and $pq = 5a$. Then $r = qv$ and $s = -pv$ with $|v|^2 = 1/9$ and p, q odd, so π_i divides neither, multiplication by them is invertible mod π_i , and $\gamma_i(r) = 0 \iff \pi_i \mid v \iff \gamma_i(s) = 0$.

Lemma 26. *In that case π_i does not divide v .*

Proof. If $v = \pi w$ with $w \in R$ then, π being real, $|v|^2 = \pi^2|w|^2$, so $|w|^2 = 1/(9\pi^2) = (58 \mp 10\sqrt{33})/144$ and (3) demands $A^2 + 33B^2 + 3C^2 + 11D^2 = 58$. The congruences force $A \equiv B \equiv C \equiv D \pmod{2}$: all even makes the form $\equiv 0 \pmod{4}$ against $58 \equiv 2$; all odd makes every square $\equiv 1 \pmod{8}$, so the form is $\equiv 1 + 33 + 3 + 11 = 48 \equiv 0 \pmod{8}$, again against $58 \equiv 2$. Neither is possible. $\square$

So both colours are equal and non-zero, and the vector imposes the same diagonal condition (c, c) that the shell displacements of Step 2 impose. Nothing new. (Two shorter routes reach the same place. Corollary 14 gives $c \neq 0$ at once, since $c = 0$ would put v in $\pi_i R$; and $c = 0$ would put v in $2R$, forcing $|v/2|^2 = 1/36$, which Section 4 records as unattained—congruence (ii) refuses its only candidate, $(\pm 2, 0, 0, 0)$. That second route is the one the accompanying code takes. Lemma 26 needs neither, and proves the stronger statement that neither prime divides v .)

Positive λ , $\alpha, \beta \geq 1$. This sign occurs and is not covered by the bound above: $\alpha\beta \geq 1$ now, so the rearrangement gives $a[m(\alpha + \beta)^2 - 1] = m\alpha\beta$, and $4a > 1$ —which $\sqrt{4a - 1}$ needs—is exactly $m(\alpha - \beta)^2 < 1$, so $(\alpha - \beta)^2 < 9$ again. The dichotomy is parity, not size. If α, β have opposite parity, one is even, both primes divide the corresponding one of r, s , and the other colour is again non-zero by Corollary 14. If both are odd then $\gamma_i(r) = \gamma_i(s) = \gamma_i(v)$ and the pair is the diagonal (c, c) of Step 2, with $c \neq 0$ by Corollary 14. Nothing new either way. The case is not vacuous: $a = 7/19$ is odd and of rank 8, and $v = \frac{5}{6} + \frac{i\sqrt{3}}{6}$, of squared norm $7/9$, gives a cross unit vector with $\lambda = 1$.

(b) λ irrational. Then $|v|^2$ is irrational and the bound of (a) is unavailable; the three cases above need none of it. For the record, irrational λ occurs only at even a in every case computed ($a = 4, 10, 16, 28$, over all rank-8 spindles to $a = 80$), but nothing rests on that now: it was the last computed-only step in the proof of Theorem 1, and the trichotomy removes it.

What the sweep checks. The accompanying code checks Step 4 and the analysis above against the actual cross unit vectors of every rank-8 spindle to $a = 80$, with zero violations, and both halves of Corollary 14 on all 552 cross vectors with $\lambda \neq -1$ to $a = 60$. Each prime divides r or s in every one of them—not only at even a , where the corollary asks for it, but at odd a as well—with exactly two exceptions, $a = 39$ and $a = 51$. Both are the $|d| = 2$ case above, where p and q are odd and neither r nor s is divisible; there Lemma 26, or the trichotomy, applies instead. The trichotomy is checked

on the same 552 directly: where $4 \nmid a$, no prime divides *both* r and s , again with zero counterexamples.

Remark 27. An earlier form of this analysis asserted that every rational λ is a ratio of *consecutive* integers, $|p - q| = 1$, measured over every rank-8 spindle with $a \leq 35$. That is false: at $a = 39$ the cross vectors have $\lambda = -13/15$ and $-15/13$. The measurement had stopped four short of its own counterexample. The bound $|d| \leq 2$ above is a proof, and $|d| = 2$ occurs exactly when $pq = 5a$ with $m = 1/9$ —which to $a = 80$ means $a = 39$ and $a = 51$, the two the sweep singles out, since $a = p(p + 2)/5$ gives rank 4 at $p = 3$ and 5 and lands past 80 at $p = 23$. The positive- λ branch refutes the old assertion a second time, from the other side: $\lambda = 1$ has $|\alpha - \beta| = 0$.

CODE AND DATA

Every computation quoted here is carried out by a script in the accompanying repository, and `paper/README.md` there maps each claim in this paper to the script that reproduces it; `scripts/verify_all.sh` runs all of them in one pass. The archive contains, among the rest: the exact coordinates of all seven constructions, each file naming the field it was written in on its first line, together with the edge sets of the minimised graphs and the scripts that recompute every edge and re-prove non-4-colourability from the coordinates in a fresh process—no proof certificates are shipped, and the non-4-colourability of the four largest unions is re-established by re-running a solver rather than by checking a proof; the 66 usable cross edges of θ_4 , the 235 monochromatic patterns and the hitting-set instance behind Theorems 21–23; the two periodic colourings certifying each of the eleven exclusions of Section 7, together with the complete list of occurring residue pairs they are checked against—the global test, not the corroborating ball run—and `scripts/audit_numbers.py`, which re-derives the paper’s numbers from the definitions, sharing no code with the rest. One block of it is an exception and says so as it runs: the three lines about which empty-shell spindles admit a homomorphism colouring are read from the repository’s own classification output, not recomputed, because that file *is* the certificate search behind them.

The repository is <https://github.com/calinnedelcu/joining-rotations-artifacts>, and the snapshot accompanying this draft is the release tagged `v1.0` there; the archived DOI below pins it immutably. Its `README.md` opens with a coverage table saying which of this paper’s claims the archive settles and which it does not—among the latter, no proof certificates are shipped and $\chi = 5$ is not established for any full union—and `PROVENANCE.md` identifies the four files taken from [6] rather than produced by these scripts.

[ARCHIVAL DOI TO BE SUPPLIED]

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